Electrical Engineering Reference — Figures Companion

Editio Unica

Stephen B. Johnson

Contents

1 EE-Book Figures Companion 3
Chapter 1: Power Engineering 3
Chapter 2: Communications Engineering 10
Chapter 3: Semiconductors 16
Chapter 4: Control Systems 22
Chapter 5: Embedded Systems 32
Chapter 6: Digital Logic 34
Chapter 7: Circuit Analysis 39
Chapter 8: Signal Processing 45
Chapter 9: Electromagnetics 50
Chapter 10: Power Electronics 55
Chapter 11: Instrumentation and Measurement 62
Chapter 12: Electric Motors 64
Chapter 13: Operational Amplifiers 66
Chapter 14: National Electrical Code (NEC) 69
Chapter 15: Networking 71
Chapter 16: Antenna Design and Principles 76
Chapter 17: Radar Systems 81
Chapter 18: Optics 83
Chapter 19: Engineering Economics 87
Appendix A: Imaginary Numbers and Phasors 88
Appendix B: Arctangent and atan2 89
Appendix C: Decibels 89
Appendix D: Unit Prefixes and SI Units 90
Appendix E: Getting Started with Python and marimo 90
Appendix F: Matrix Operations 90
Appendix G: Op-Amp Circuit Collection 90

Chapter 1

EE-Book Figures Companion

This companion document collects supplementary figures for the *Electrical Engineering Reference* by topic. Unlike the figures embedded directly in the chapters, entries here are additional visualizations — generated from the `scripts/` marimo notebooks, or photographs and diagrams sourced from public-domain and CC0 archives — that illustrate chapter concepts without being woven into the main narrative. It is built and distributed separately from the main reference and the problem sets.

Each entry follows the same figure format used in the chapters: a numbered caption and the image. Externally sourced figures include a **Source:** line citing the origin and copyright status; no attribution is legally required for public-domain/CC0 works, but the source is noted for verifiability.

Chapter 1: Power Engineering

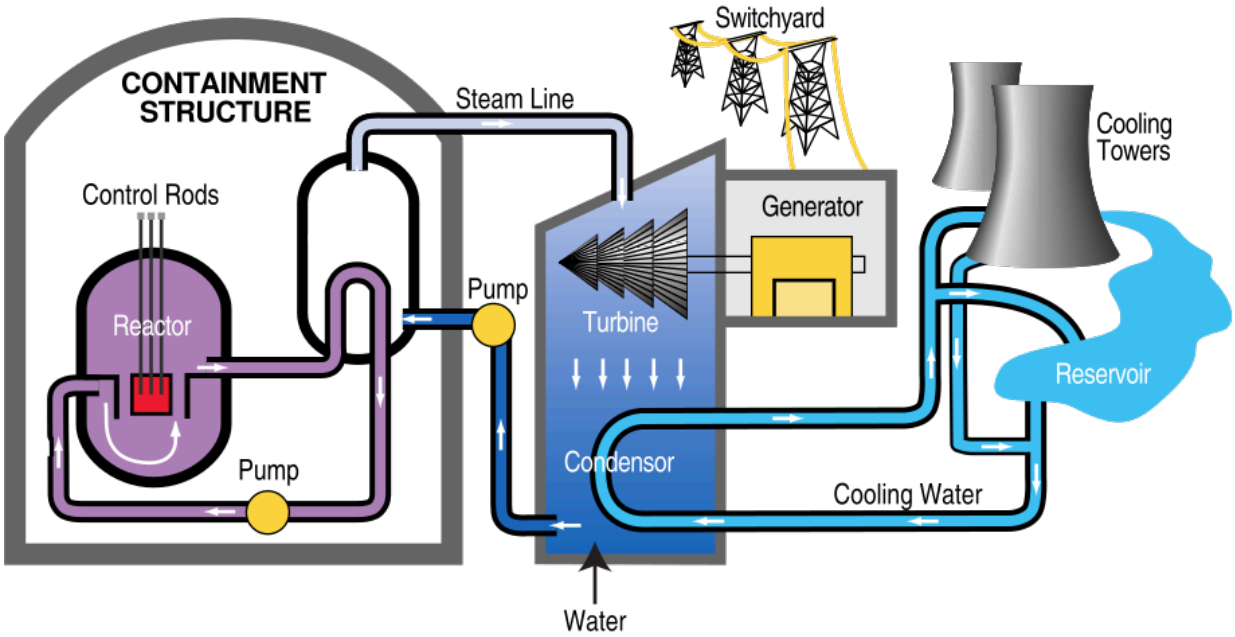

Figure C1.1: Pressurized Water Reactor (PWR) Nuclear Power Plant Diagram

Figure C1.2: High-Voltage Transmission Tower

Figure C1.3: Outdoor Power Distribution Substation

Figure C1.4: Substation Power Circuit Breaker, Metal-Clad Switchgear

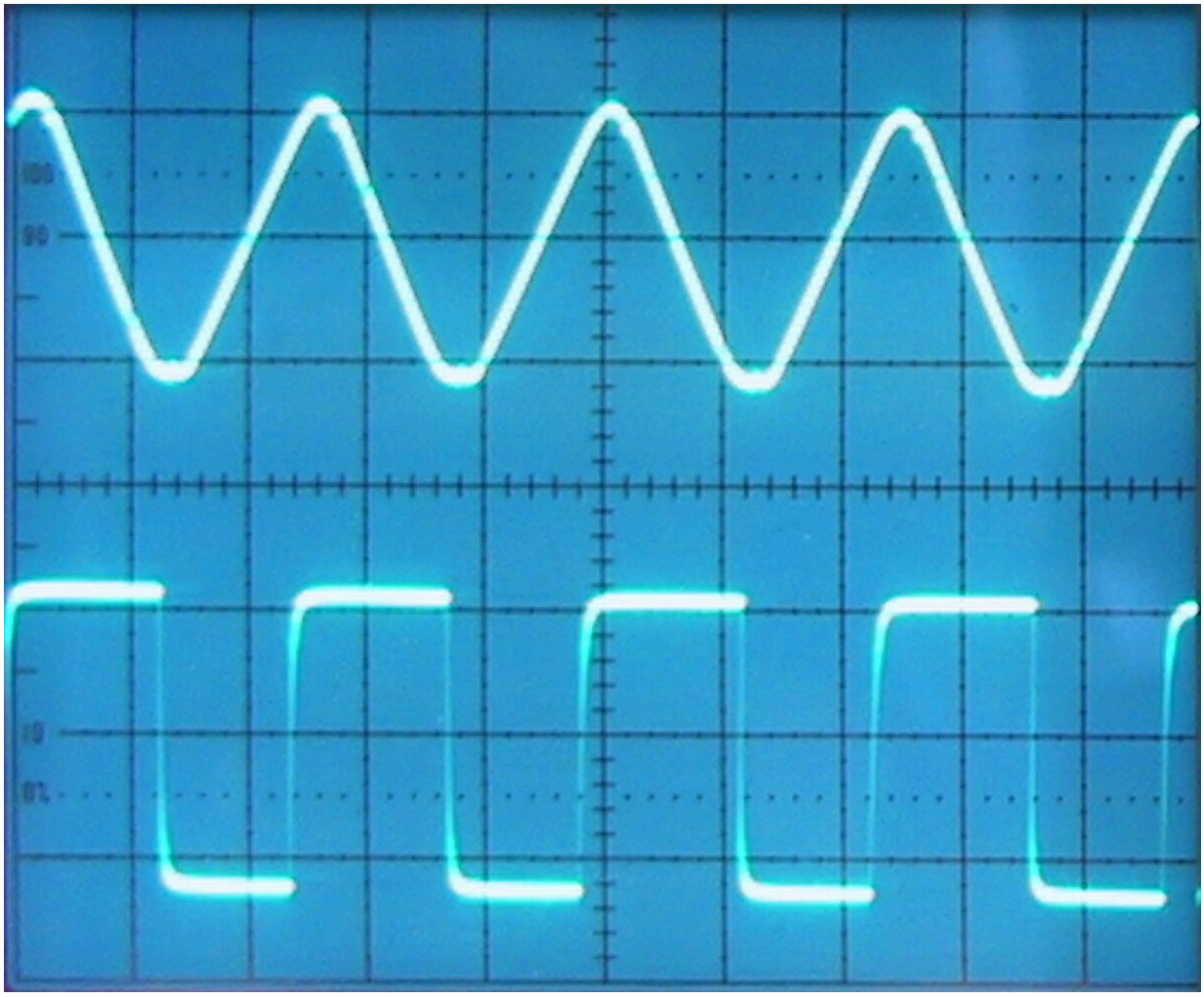

Figure C1.5: Fundamental and Harmonic Waveforms on an Oscilloscope

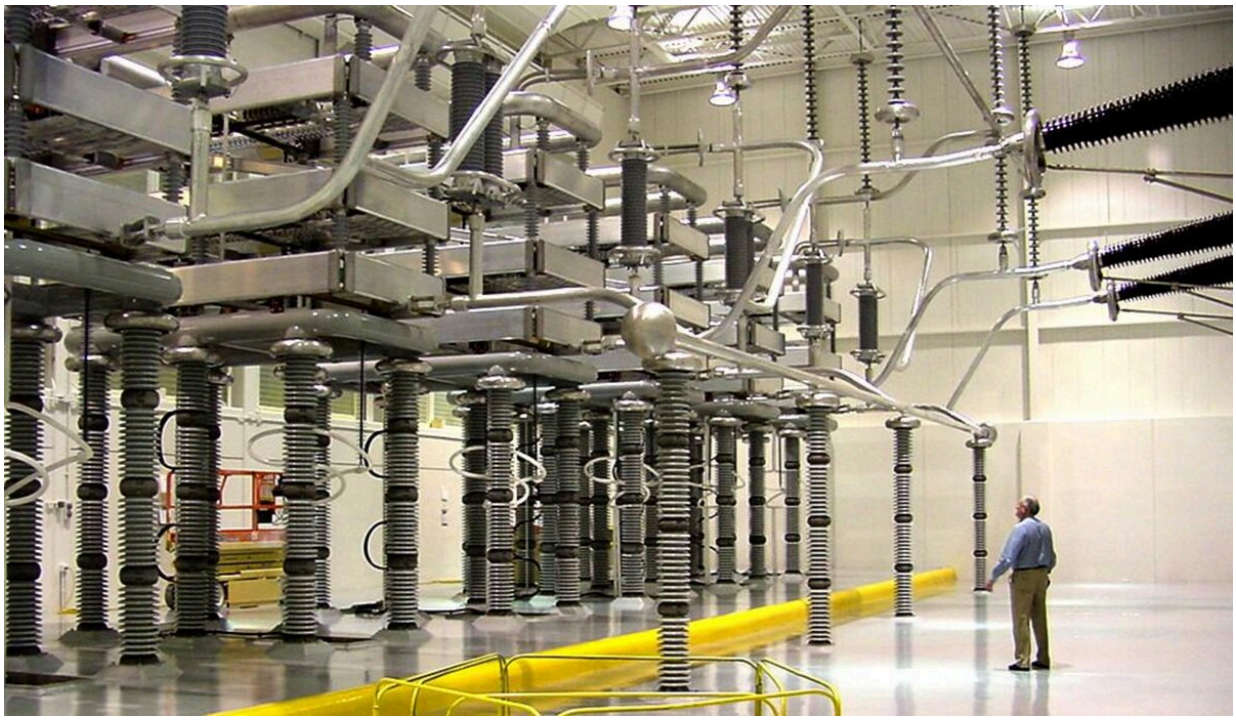

Figure C1.6: Celilo Converter Station, Pacific DC Intertie (HVDC Terminal)

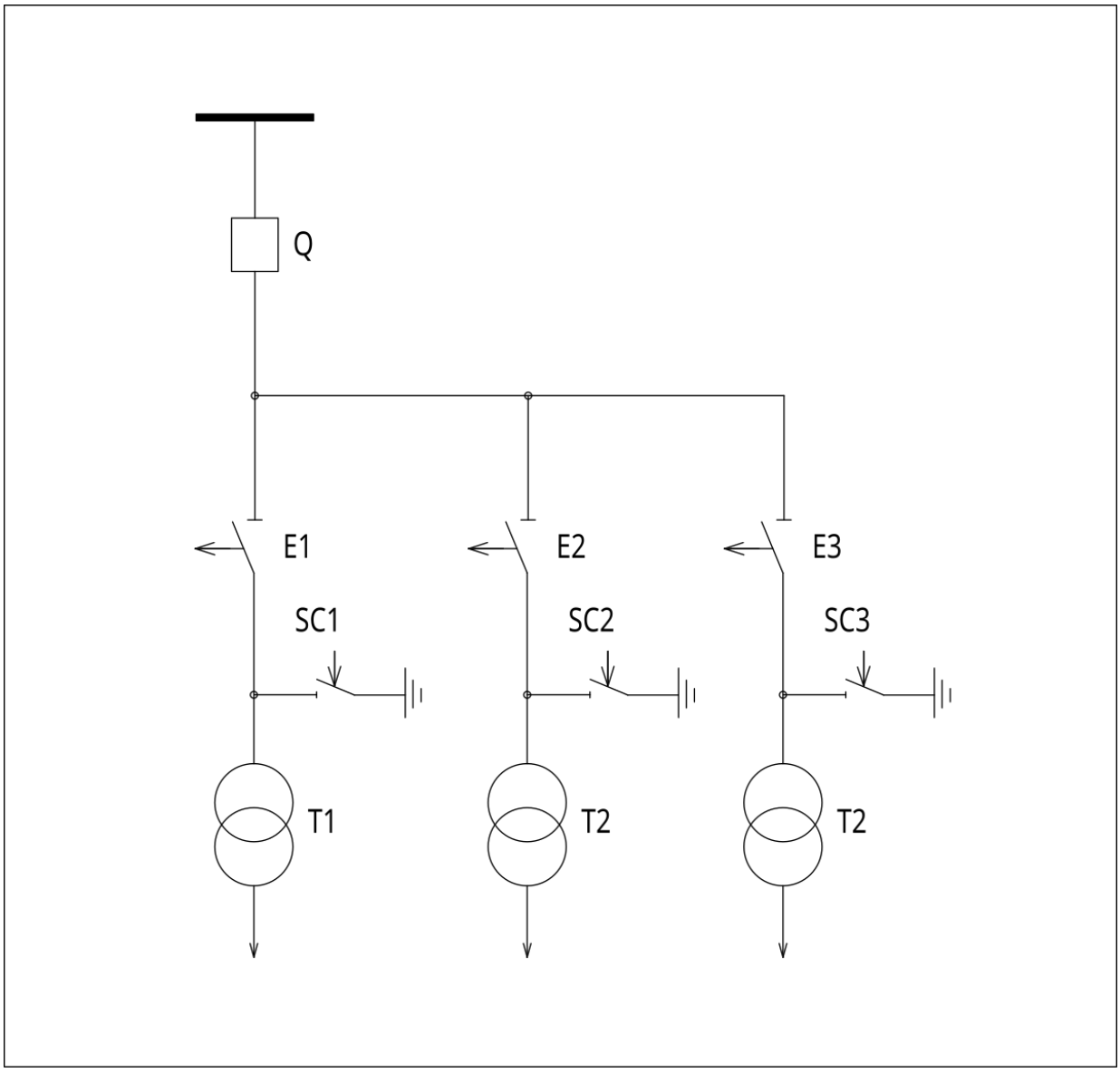

Figure C1.7: One-Line Diagram of a Distribution Substation

Figure C1.8: SCADA Operator Balancing Electrical Load

Chapter 2: Communications Engineering

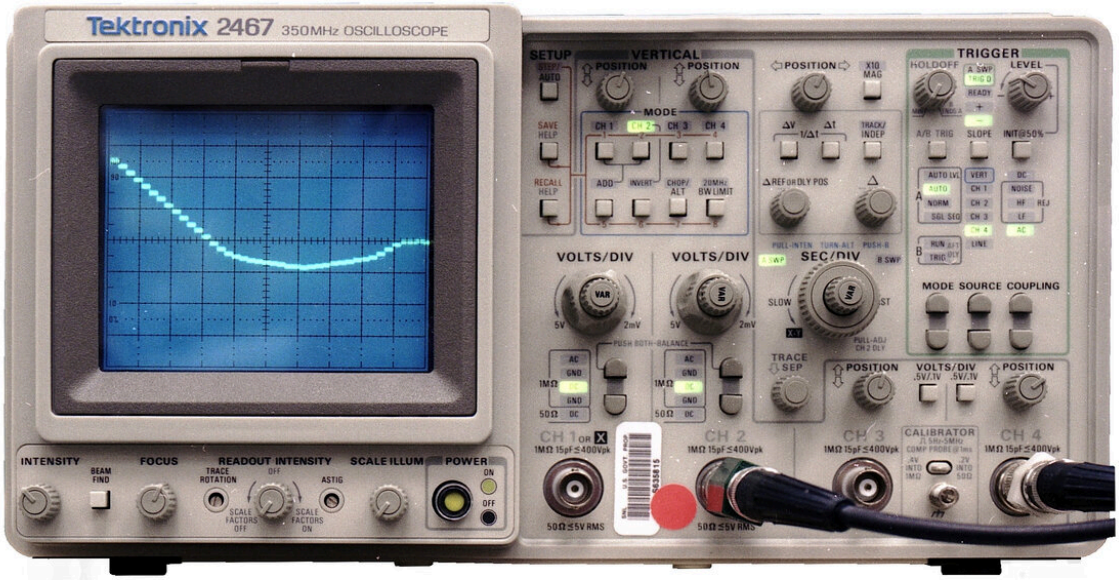

Figure C2.1: Analog Oscilloscope for Analog Signal Observation

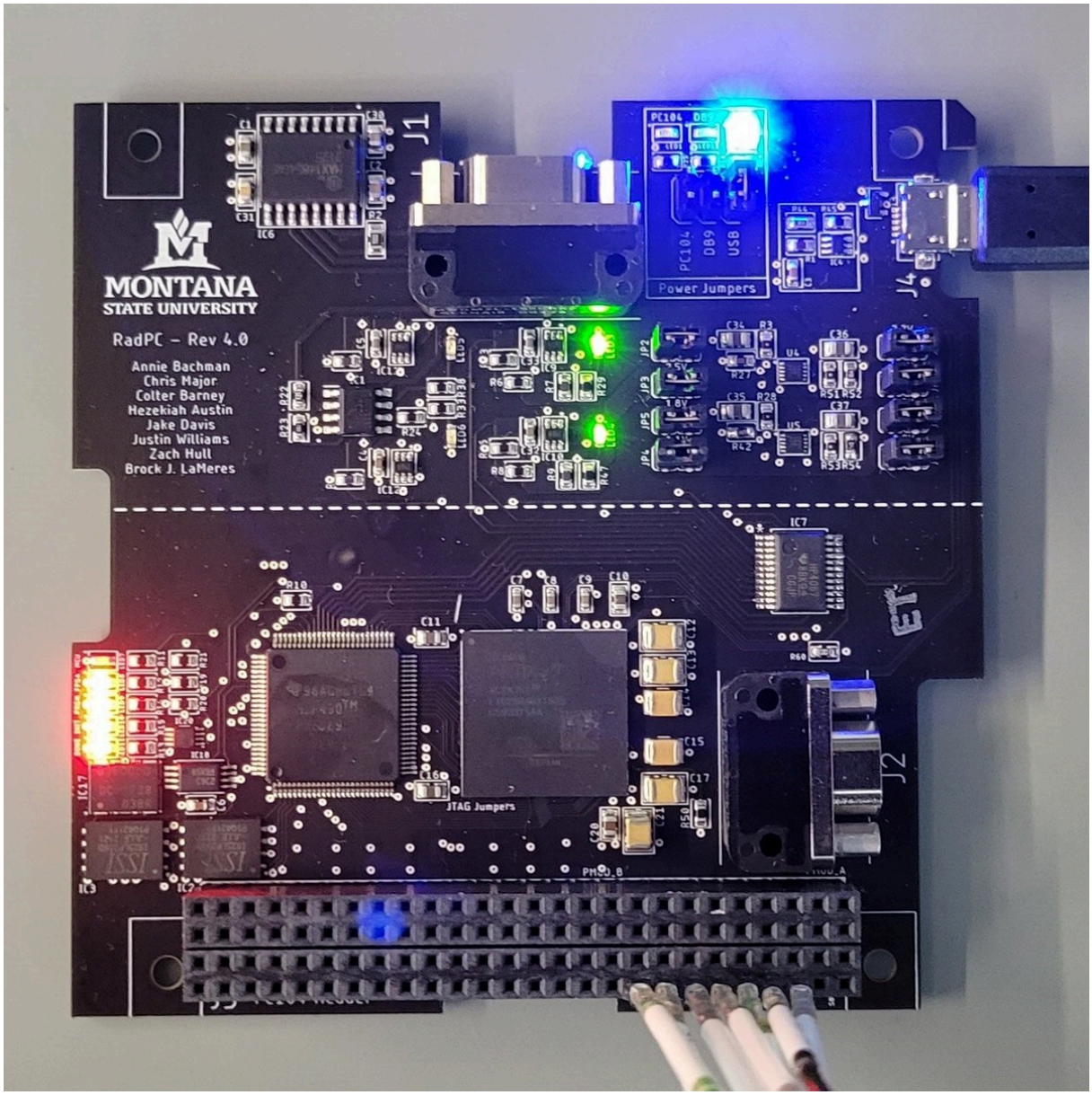

Figure C2.2: FPGA-Based Digital Signal Processing Board

Figure C2.3: HackRF One Software-Defined Radio Transceiver

Figure C2.4: DSN Deep-Space Antenna (Canberra) — Channel Coding Heritage from Deep-Space Links

Figure C2.5: Manual Telephone Switchboard, Salt Lake City, 1907

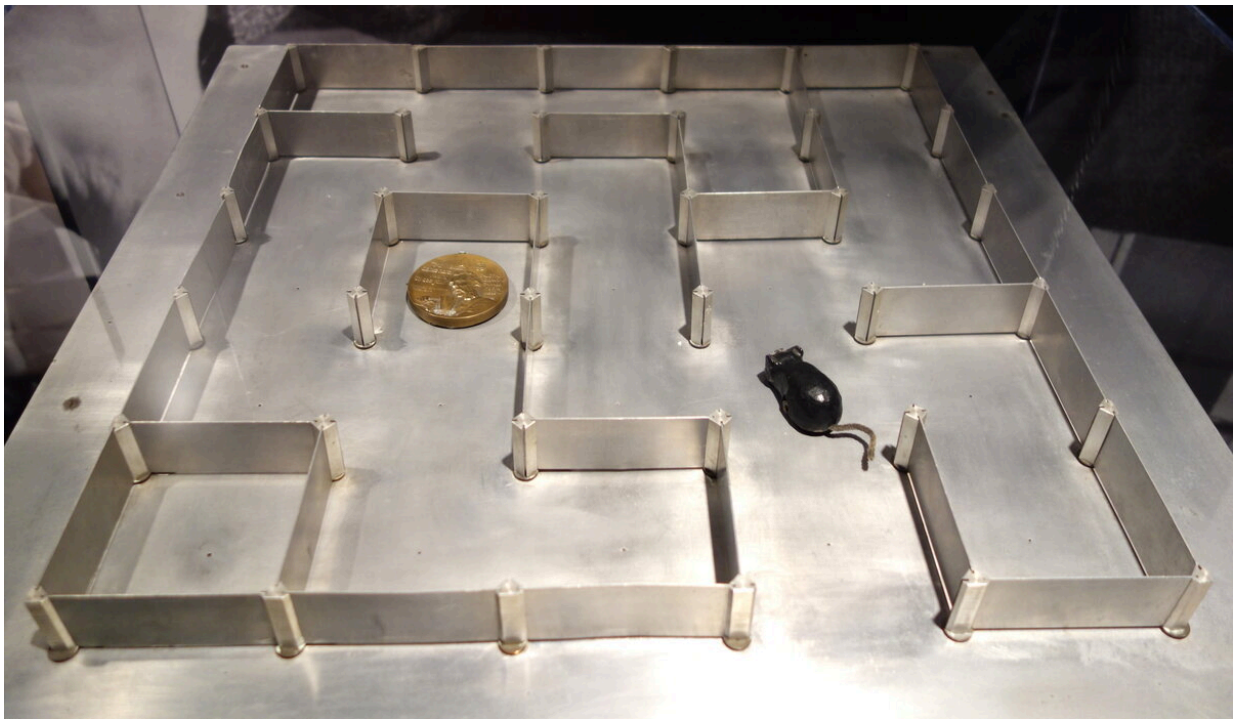

Figure C2.6: Theseus, Claude Shannon's Maze-Solving Mouse (1952)

Figure C2.7: Green Bank Radio Telescope — Low-Noise Receiver Systems

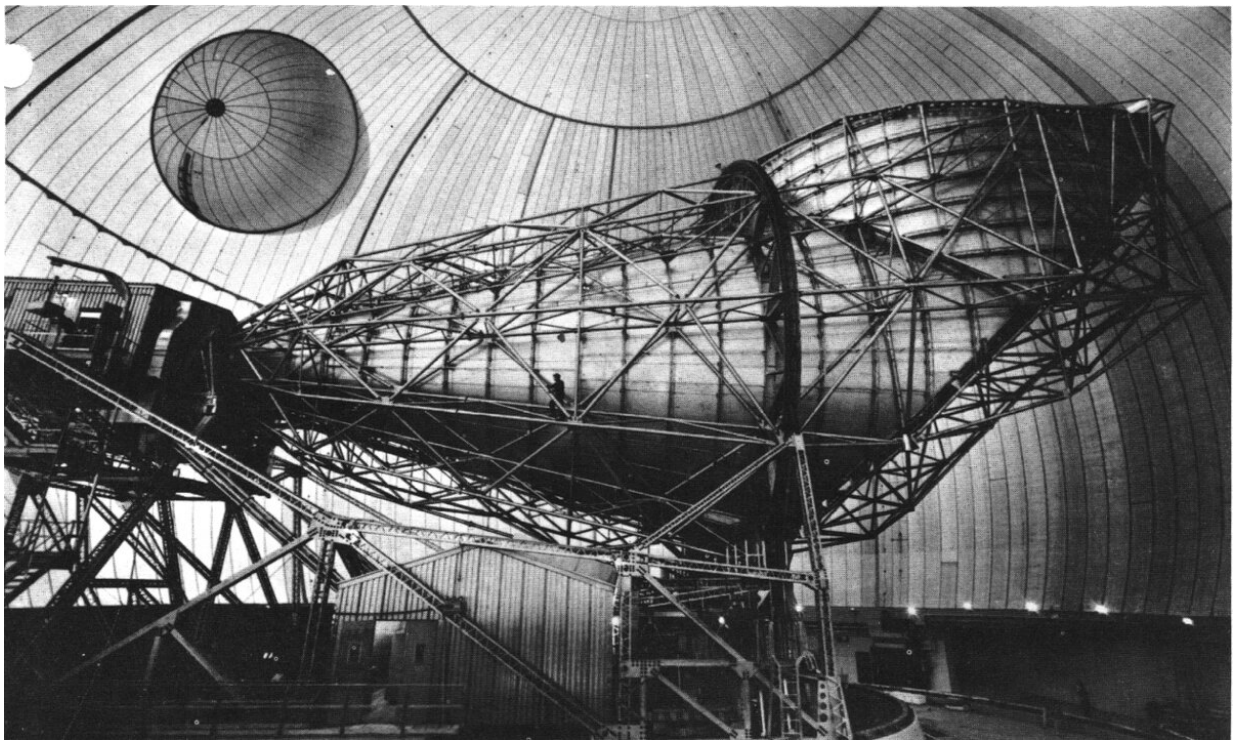

Figure C2.8: Relay 1 Ground Station Antenna — Early Satellite Communications Link Budget

Chapter 3: Semiconductors

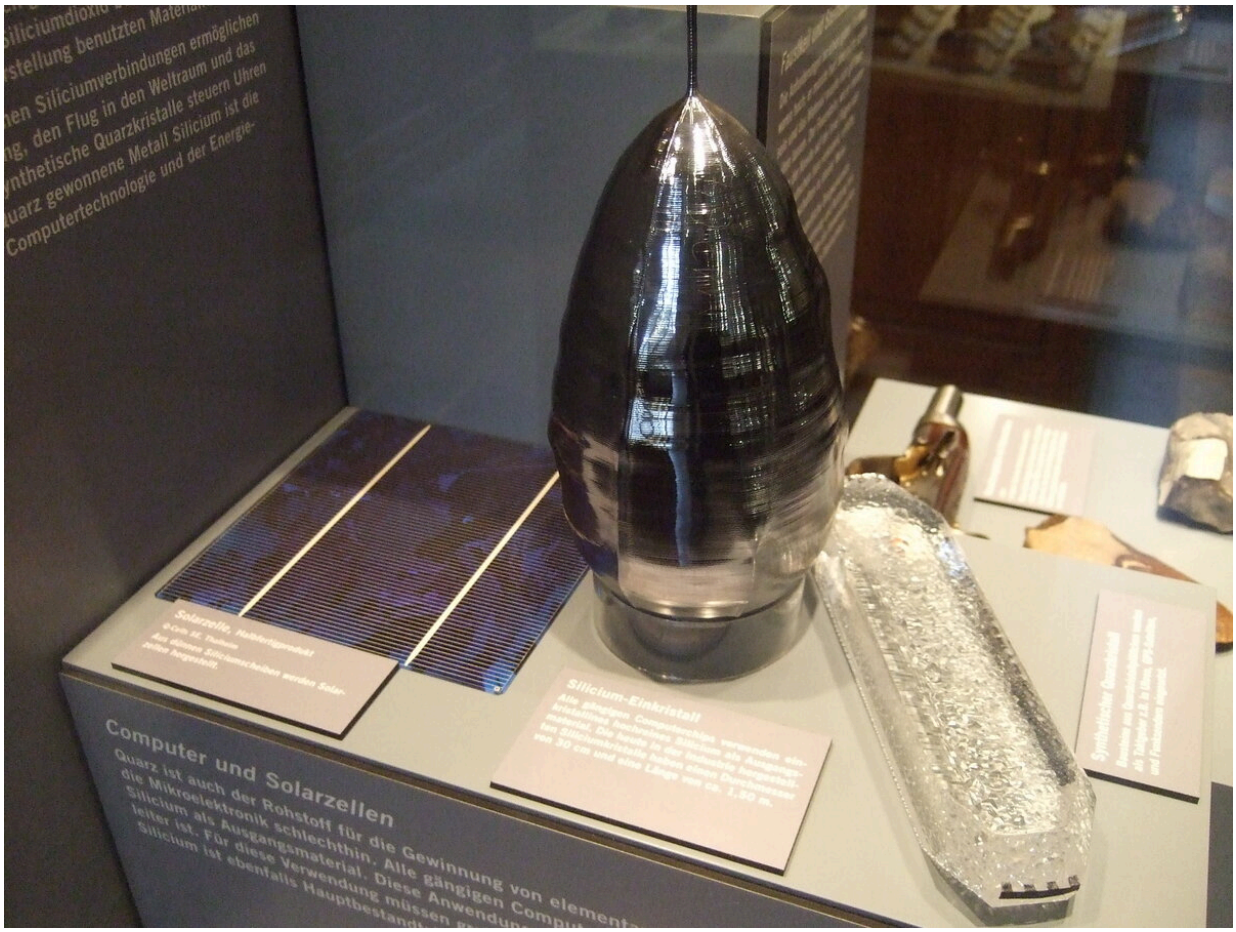

Figure C3.1: Single-Crystal Silicon Boule and Wafer

Figure C3.2: Compound-Semiconductor Laser Diode Die (Shown Against a Penny for Scale)

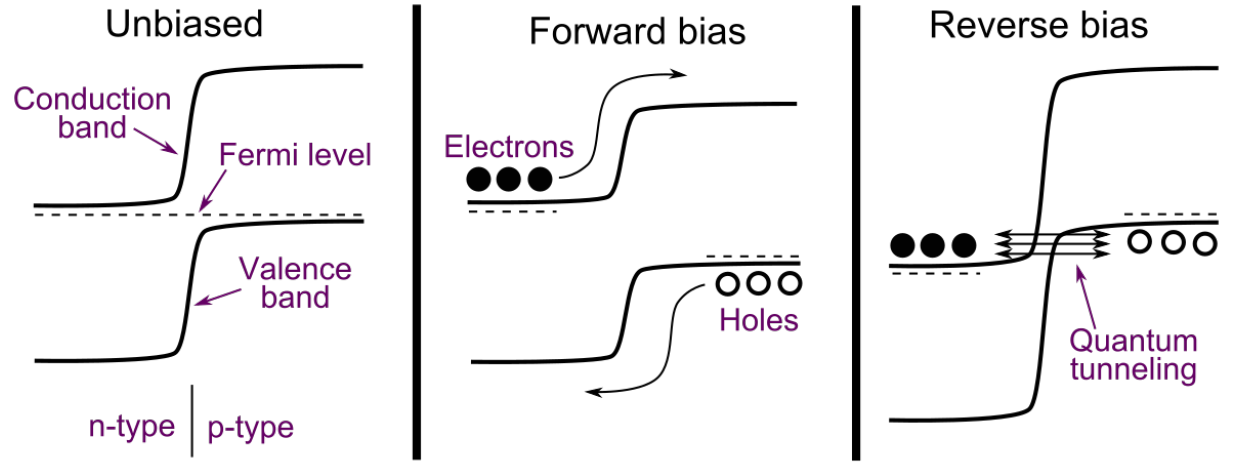

Figure C3.3: Energy-Band Diagram of a PN Junction — Unbiased, Forward-Bias, and Reverse-Bias Conditions

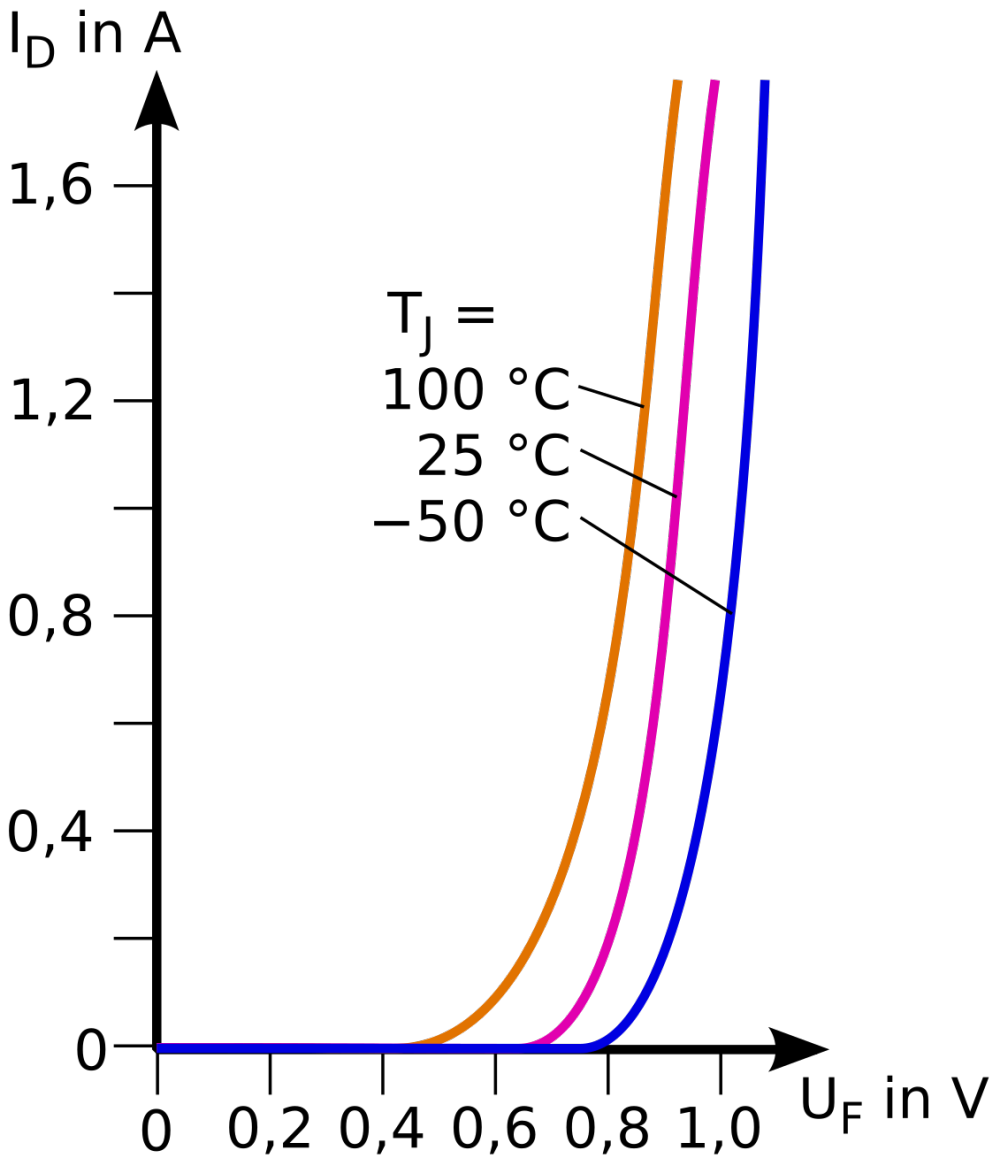

Figure C3.4: Measured I-V Characteristic Curve of a 1N4001 Diode

Figure C3.5: Replica of the First Point-Contact Transistor (Bell Labs, 1947)

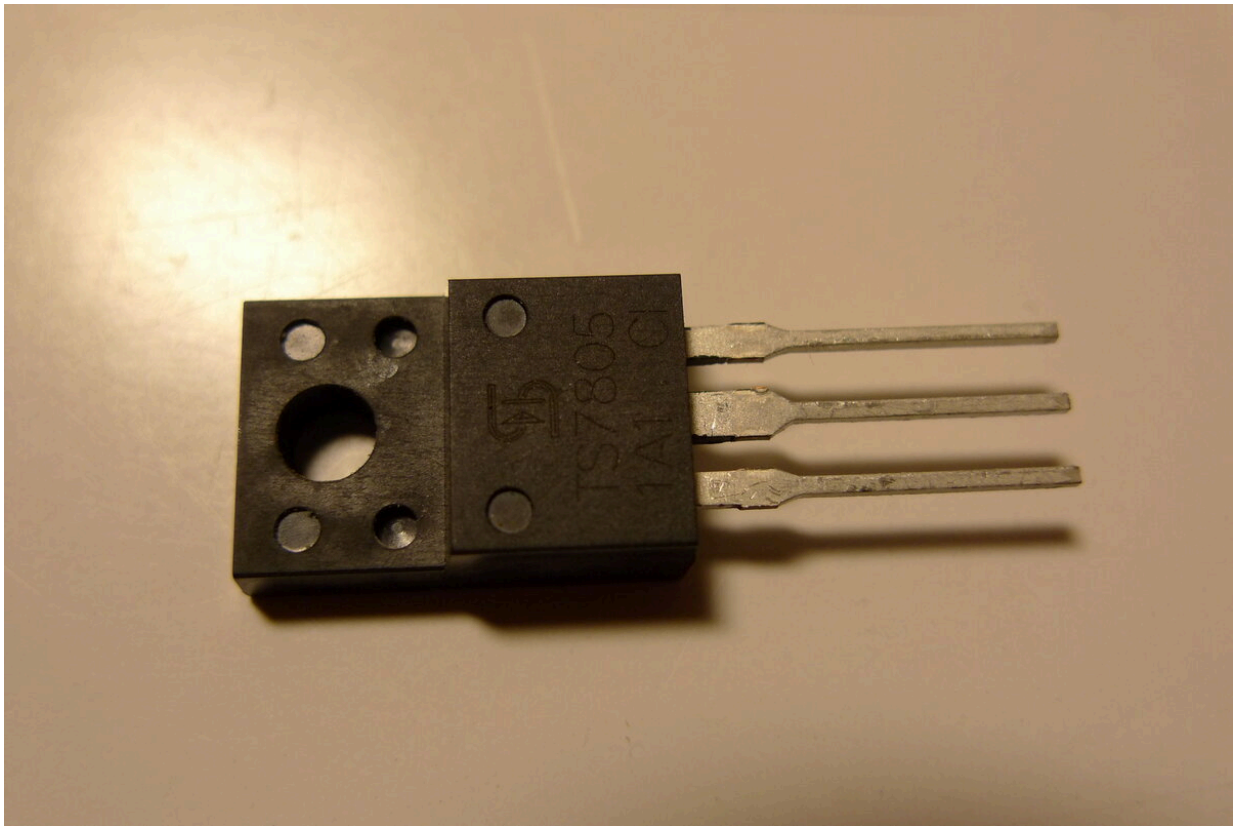

Figure C3.6: TS7805 Linear Voltage Regulator in a TO-220 Package

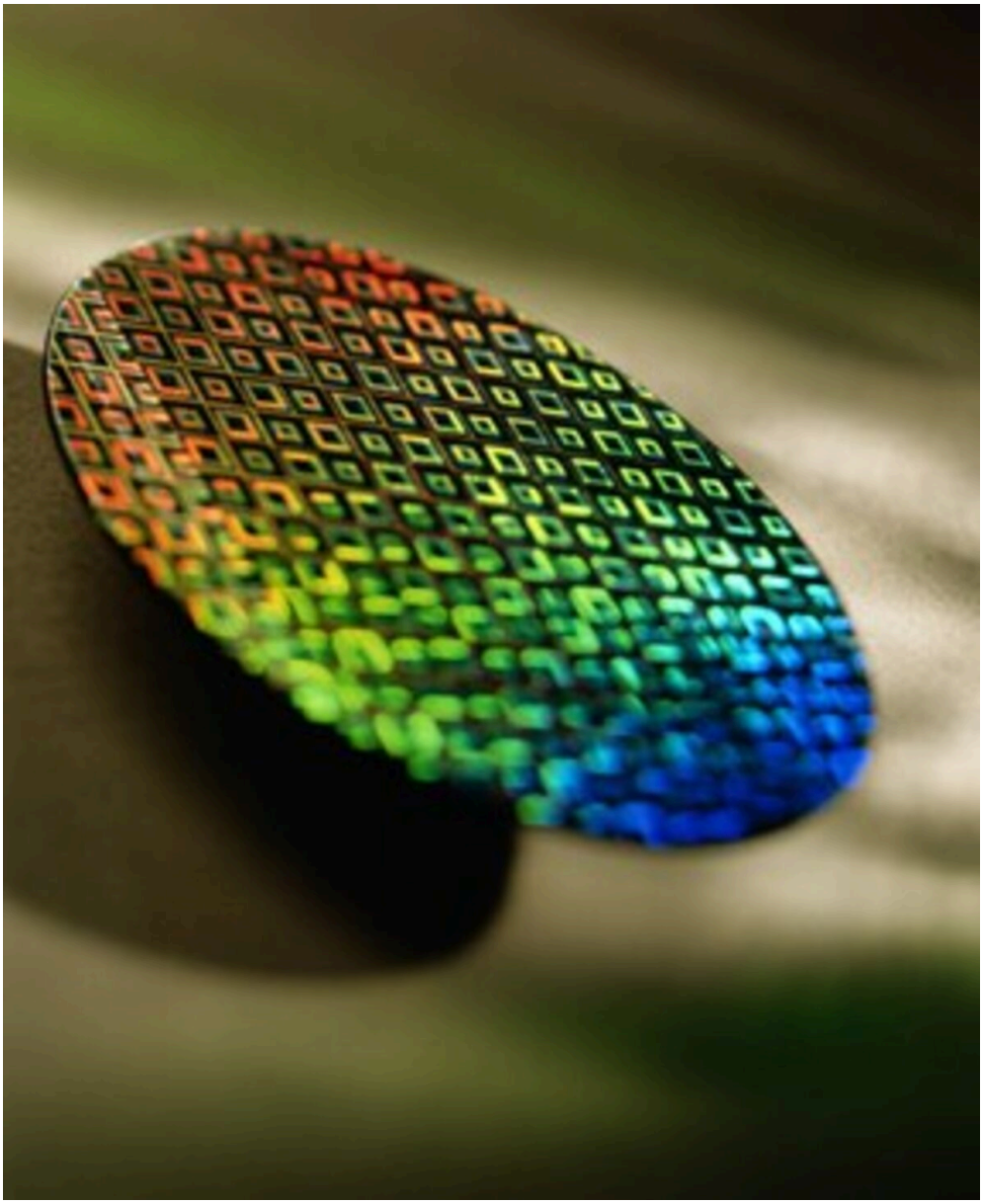

Figure C3.7: Etched Silicon Wafer Showing Individual Die

Chapter 4: Control Systems

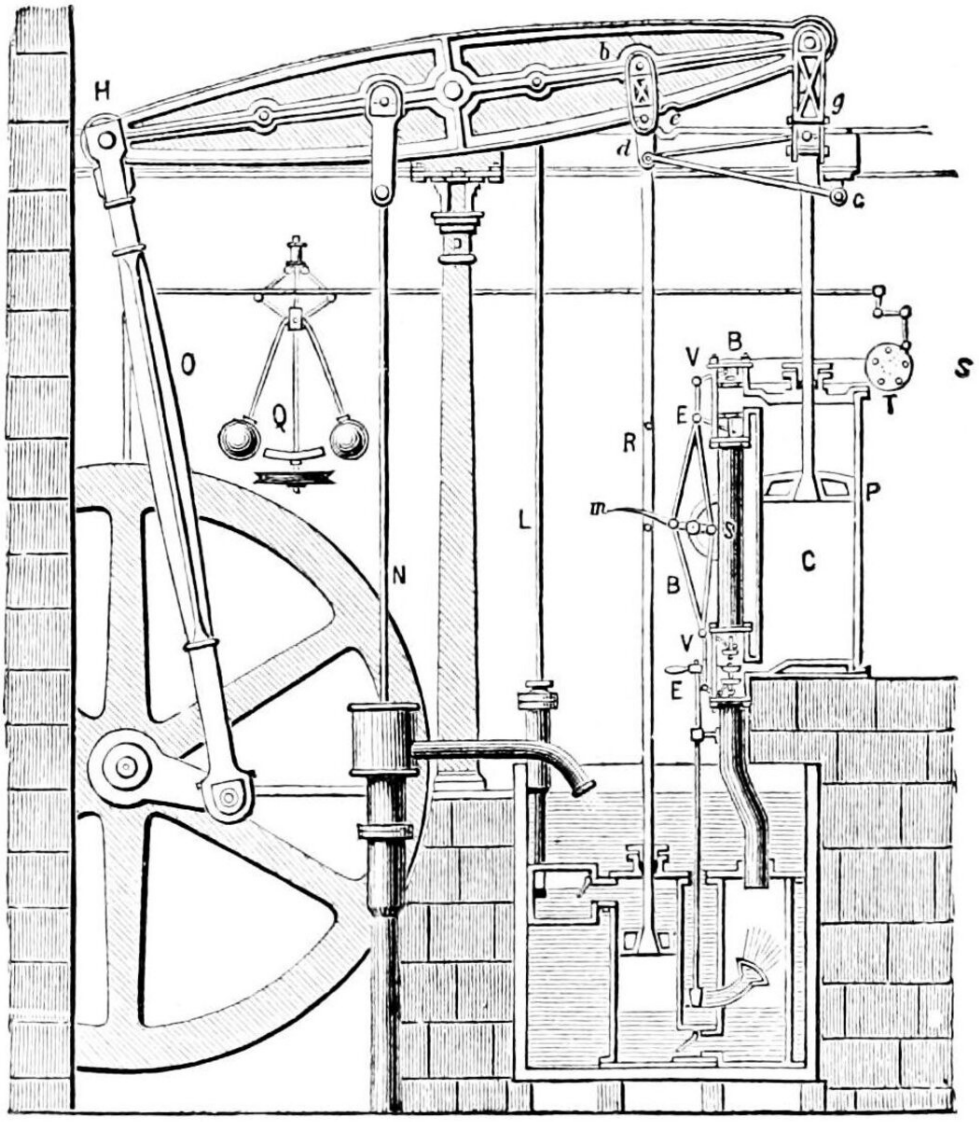

Figure C4.1: Watt Steam Engine (1780) — Uncontrolled (Open-Loop) Mechanism

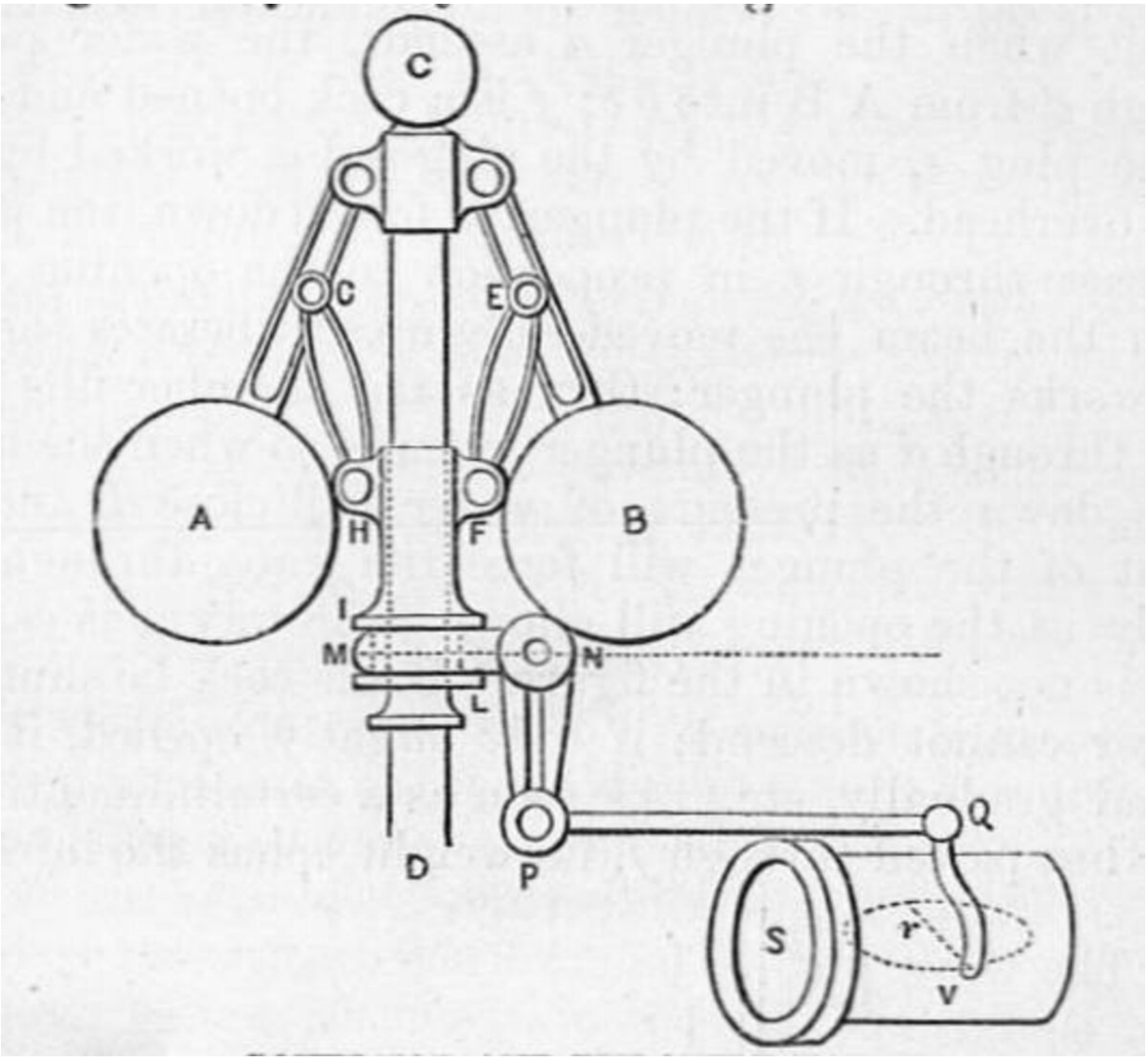

Figure C4.2: Watt's Centrifugal Flyball Governor — the First Widely Used Closed-Loop Feedback Controller

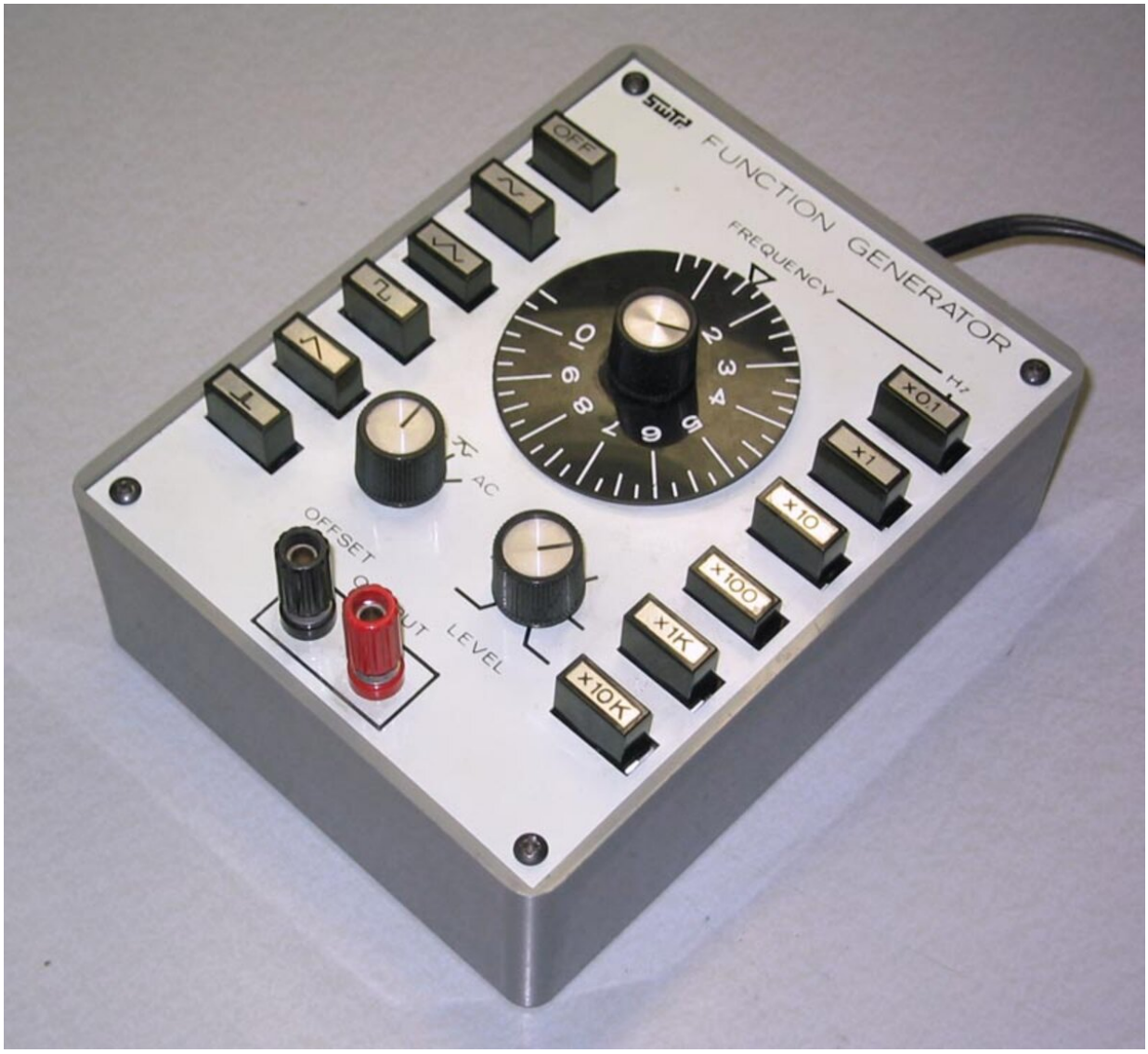

Figure C4.3: Function Generator for Producing Reference Control Signals

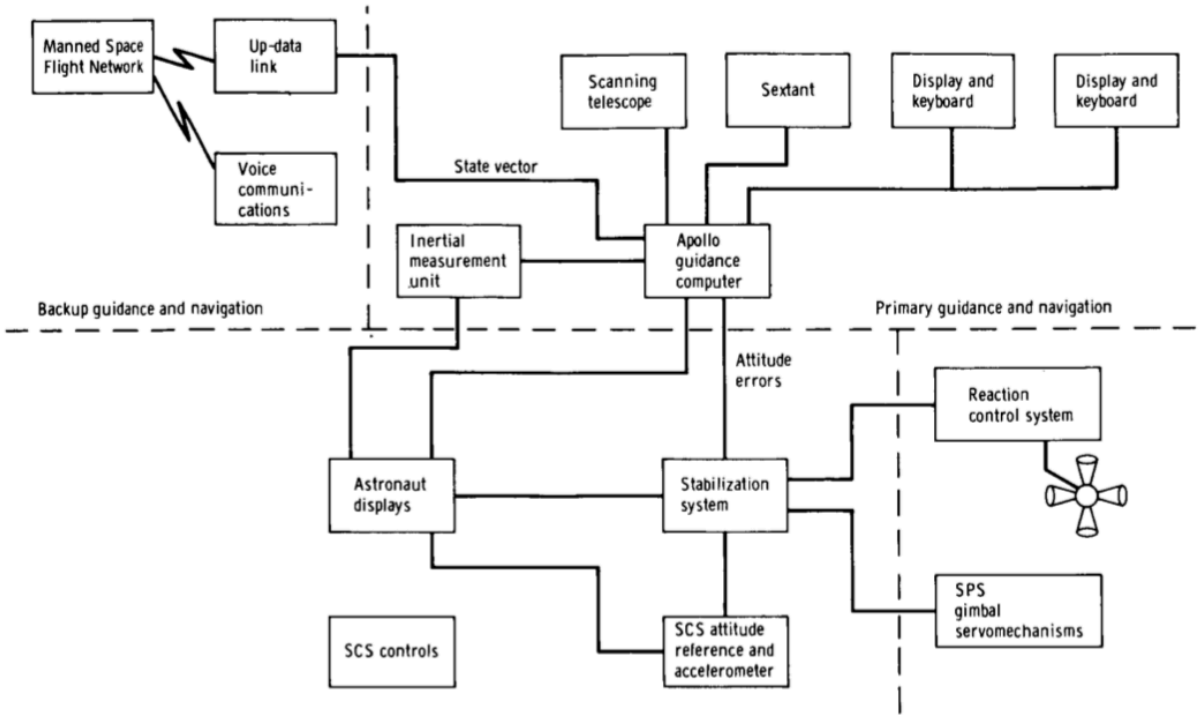

Figure C4.4: Apollo Block I Guidance System Block Diagram

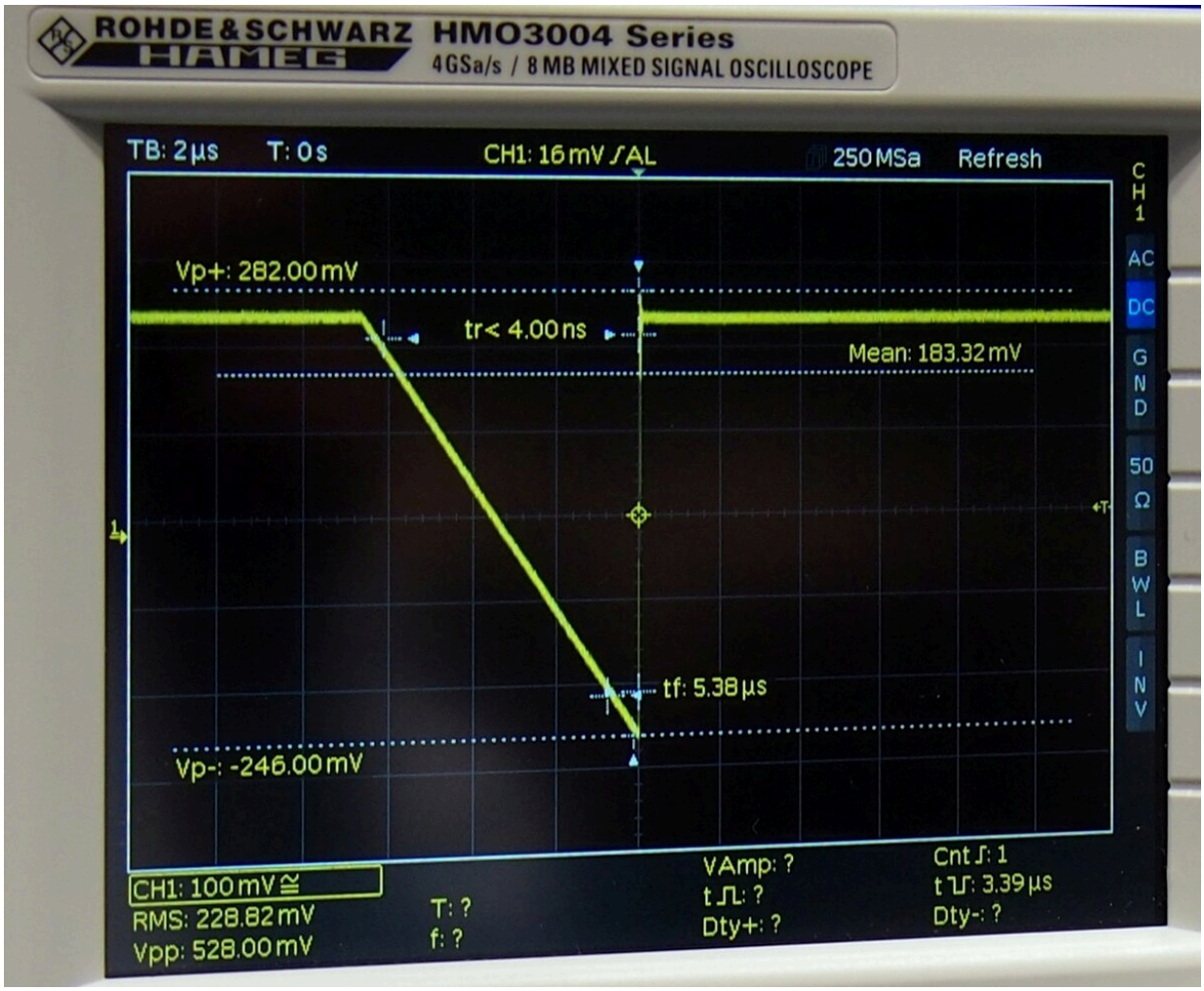

Figure C4.5: Oscilloscope Capture of a Time-Domain Step/Fall Response

Figure C4.6: Honeywell Round Thermostat (1953) — Classic Bang-Bang/PID Feedback Control

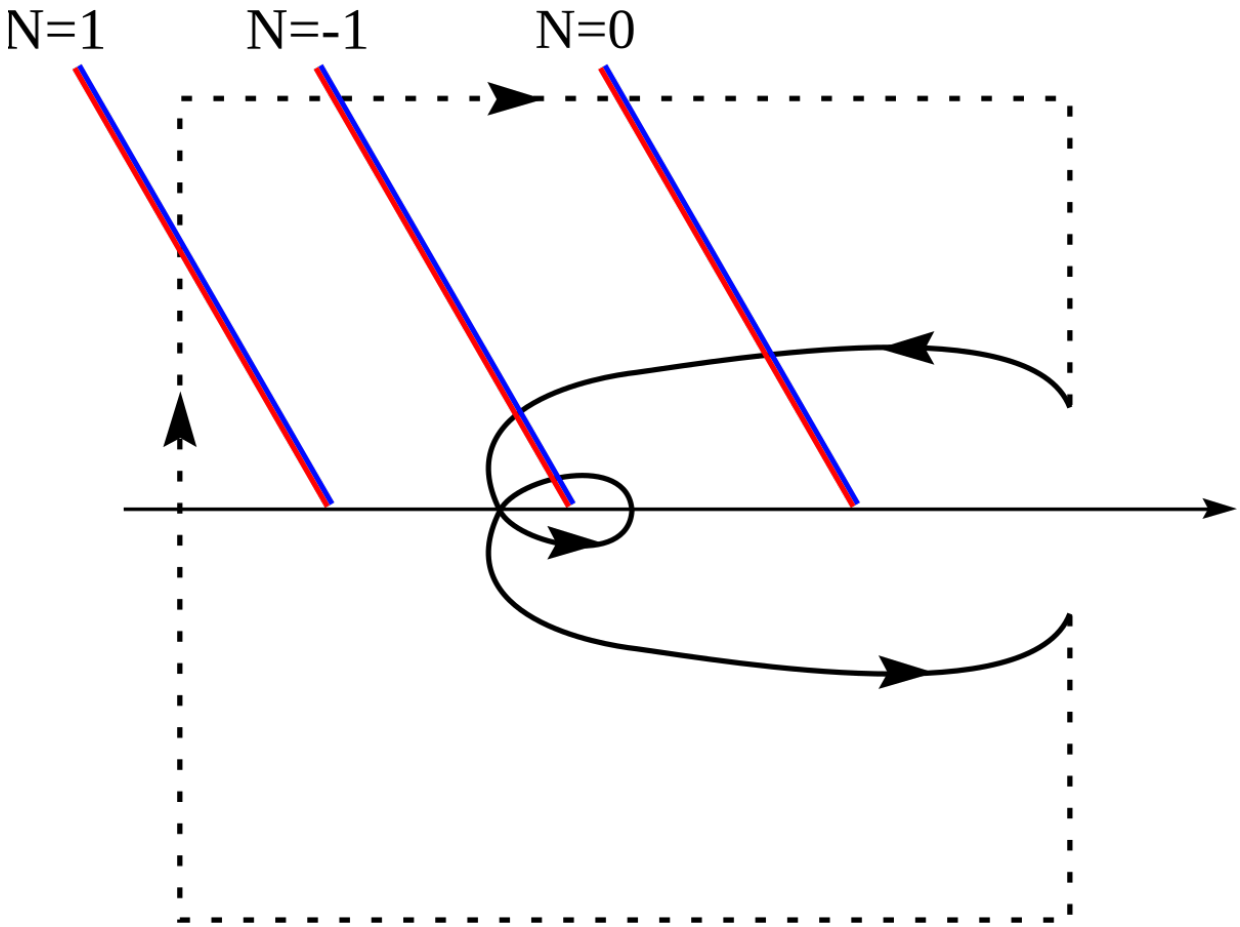

Figure C4.7: Nyquist Stability Criterion Diagram

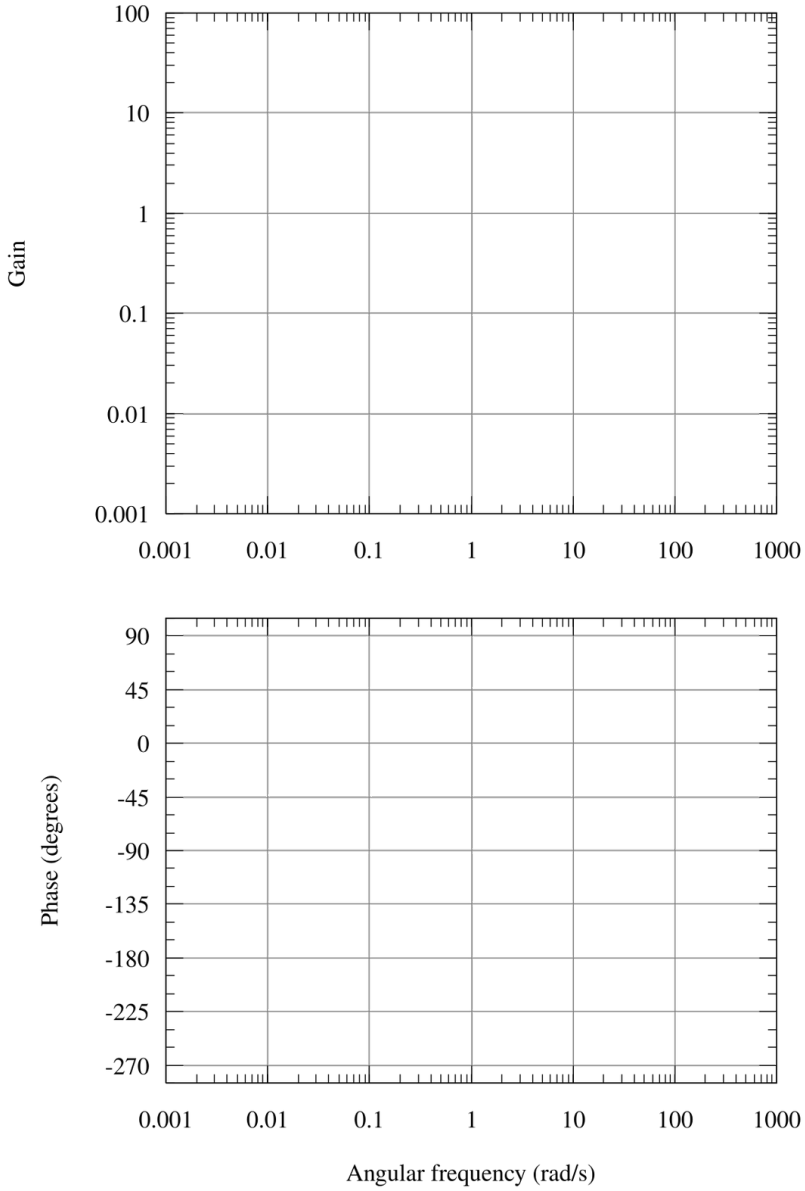

Figure C4.8: Bode Plot Template — Magnitude and Phase vs. Frequency

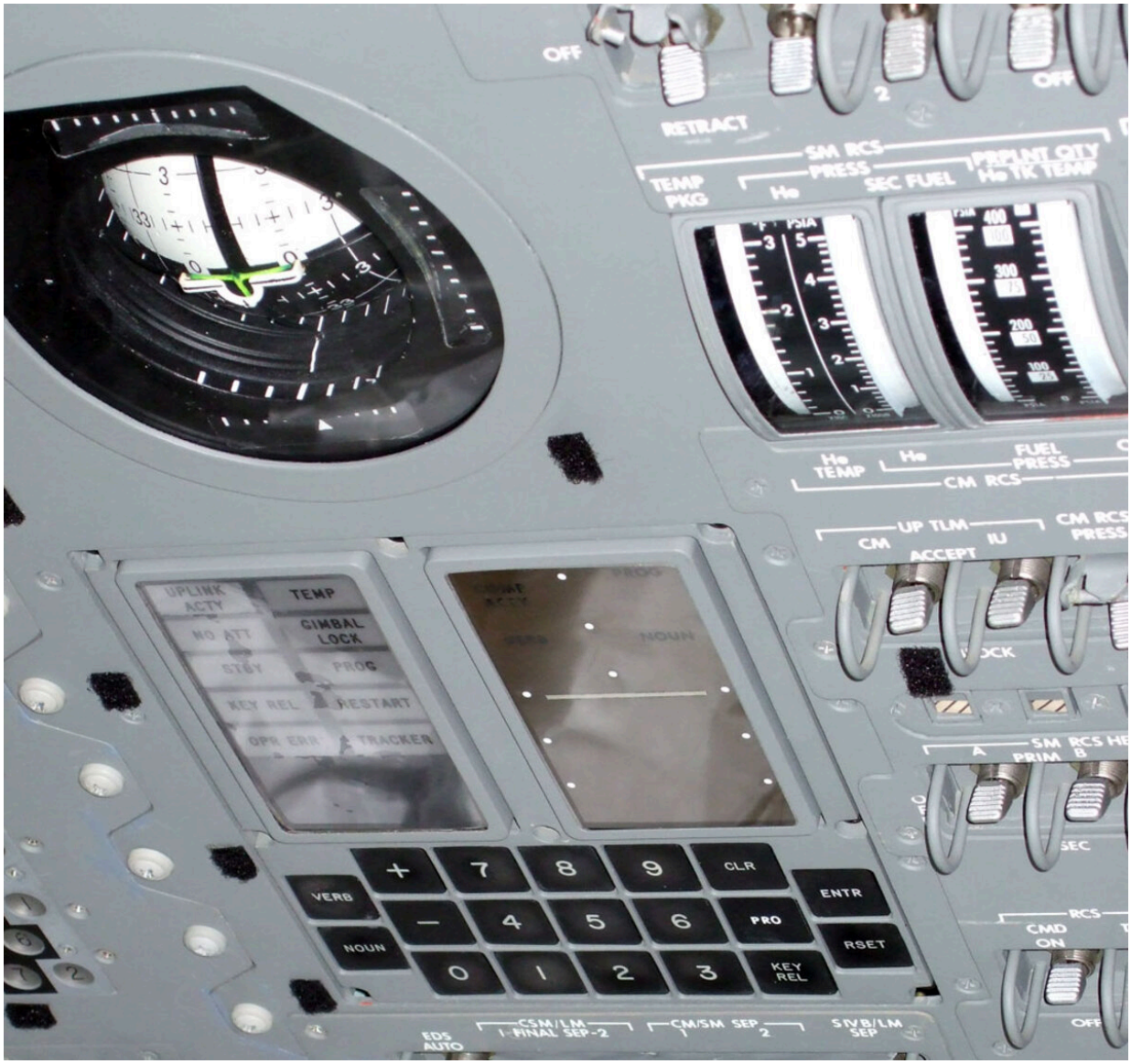

Figure C4.9: Apollo Guidance Computer DSKY (Display and Keyboard) — Onboard State Estimation and Control Interface

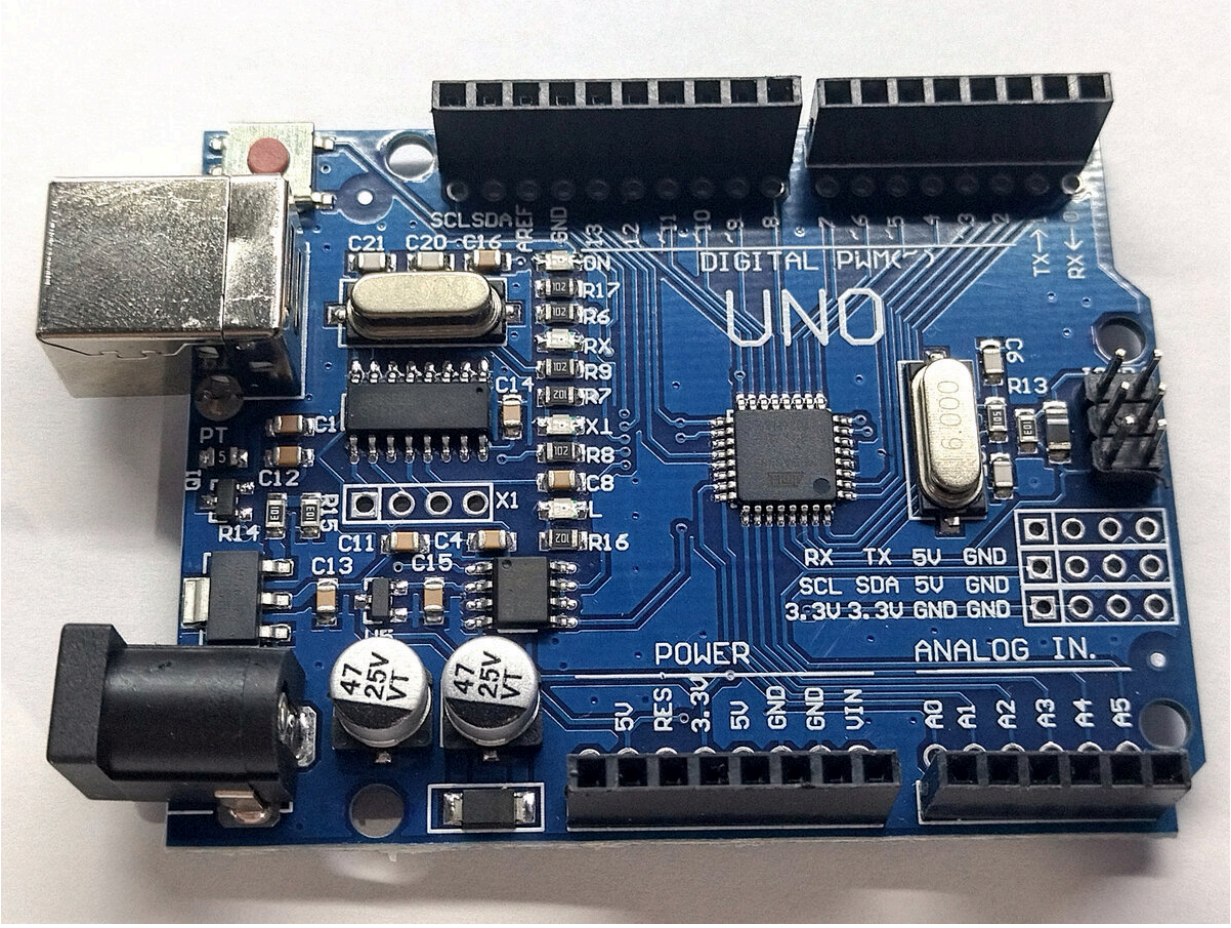

Figure C4.10: Arduino Uno Microcontroller Board — a Common Digital Control System Platform

Chapter 5: Embedded Systems

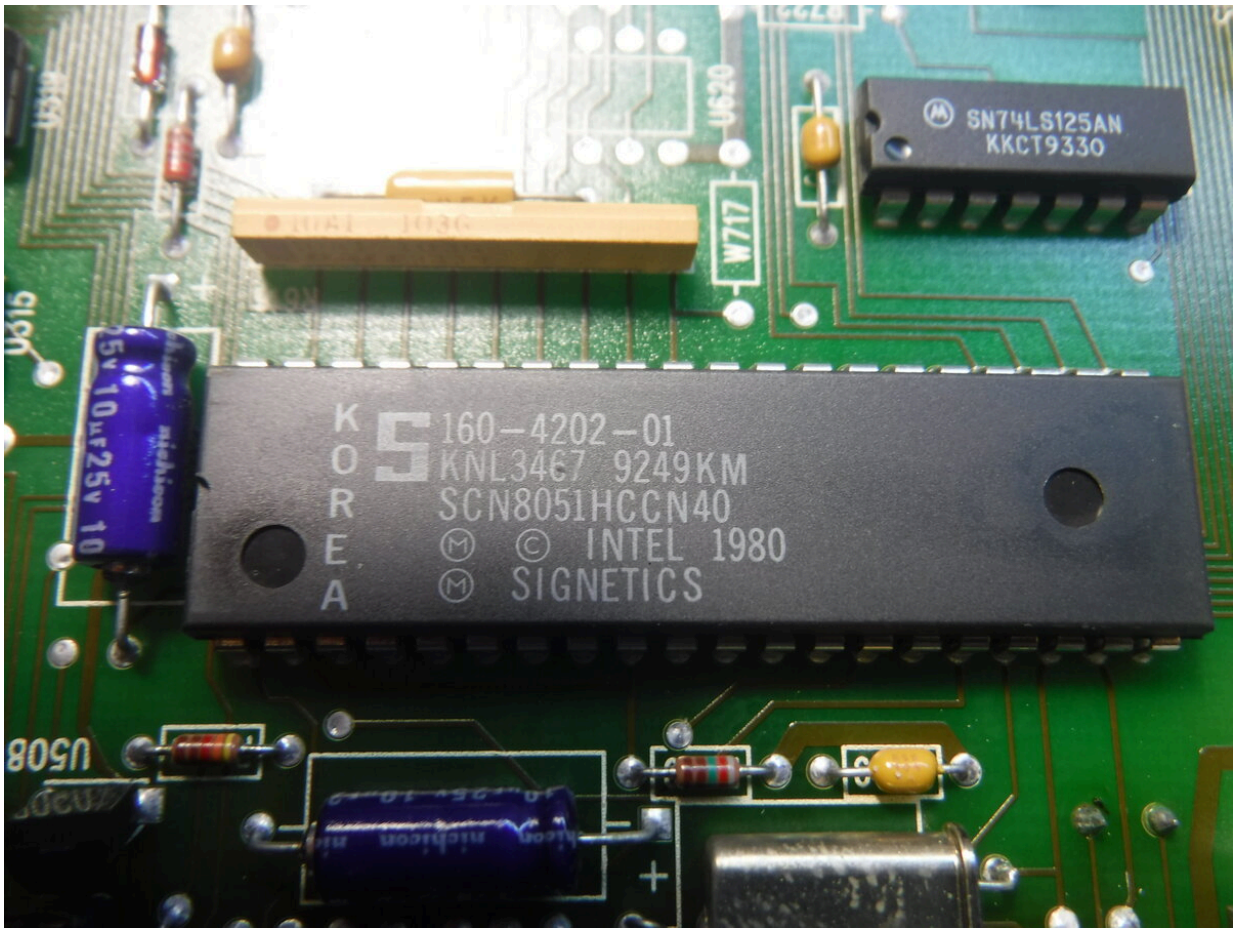

Figure C5.1: Signetics SCN8051 (Intel 8051-Compatible) Microcontroller on a Circuit Board

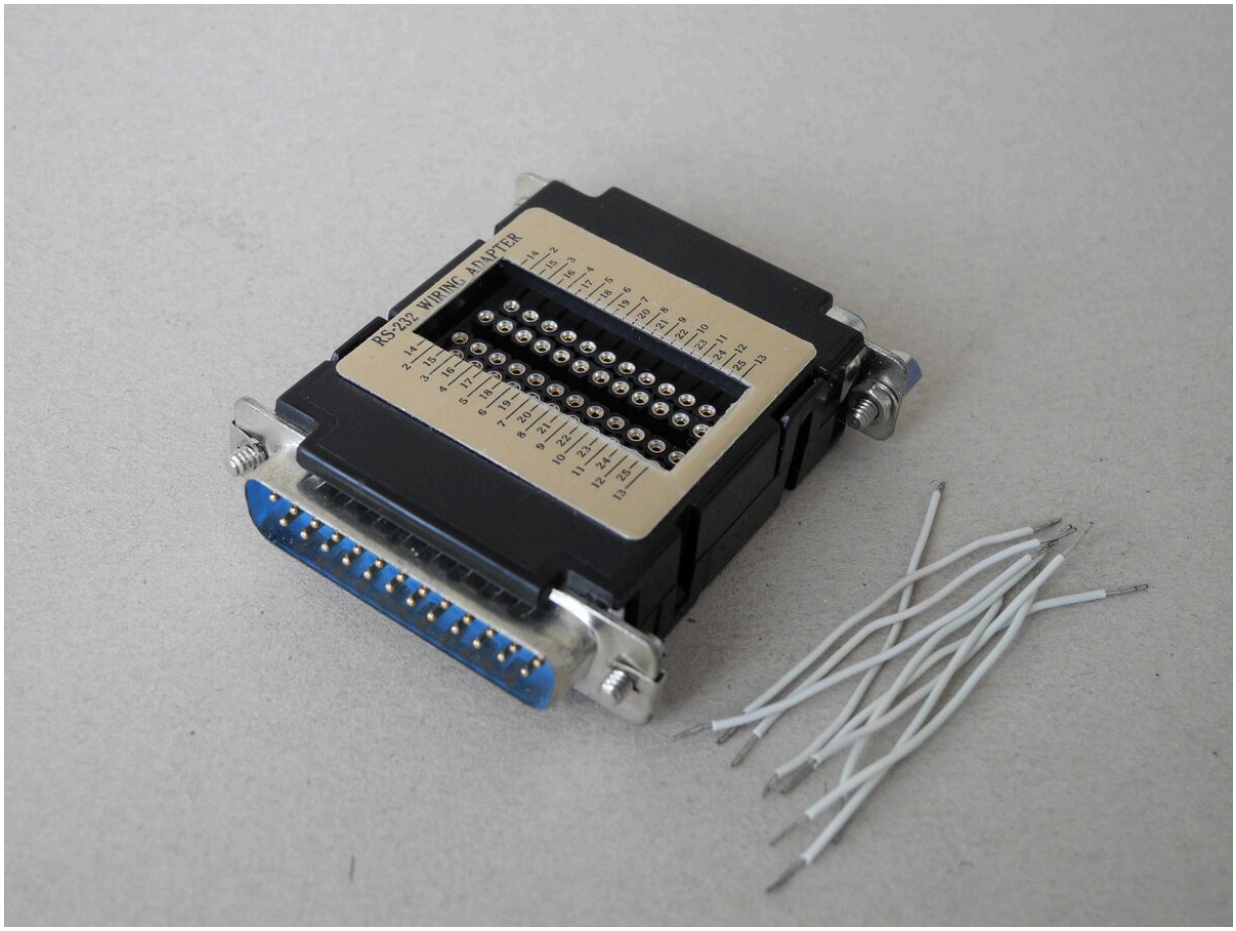

Figure C5.2: RS-232 Wiring Adapter (DB-25) for Serial Communication Interfaces

Chapter 6: Digital Logic

Figure C6.1: ERA 1103 Computer — Early Binary/Digital Calculating Machine (1955)

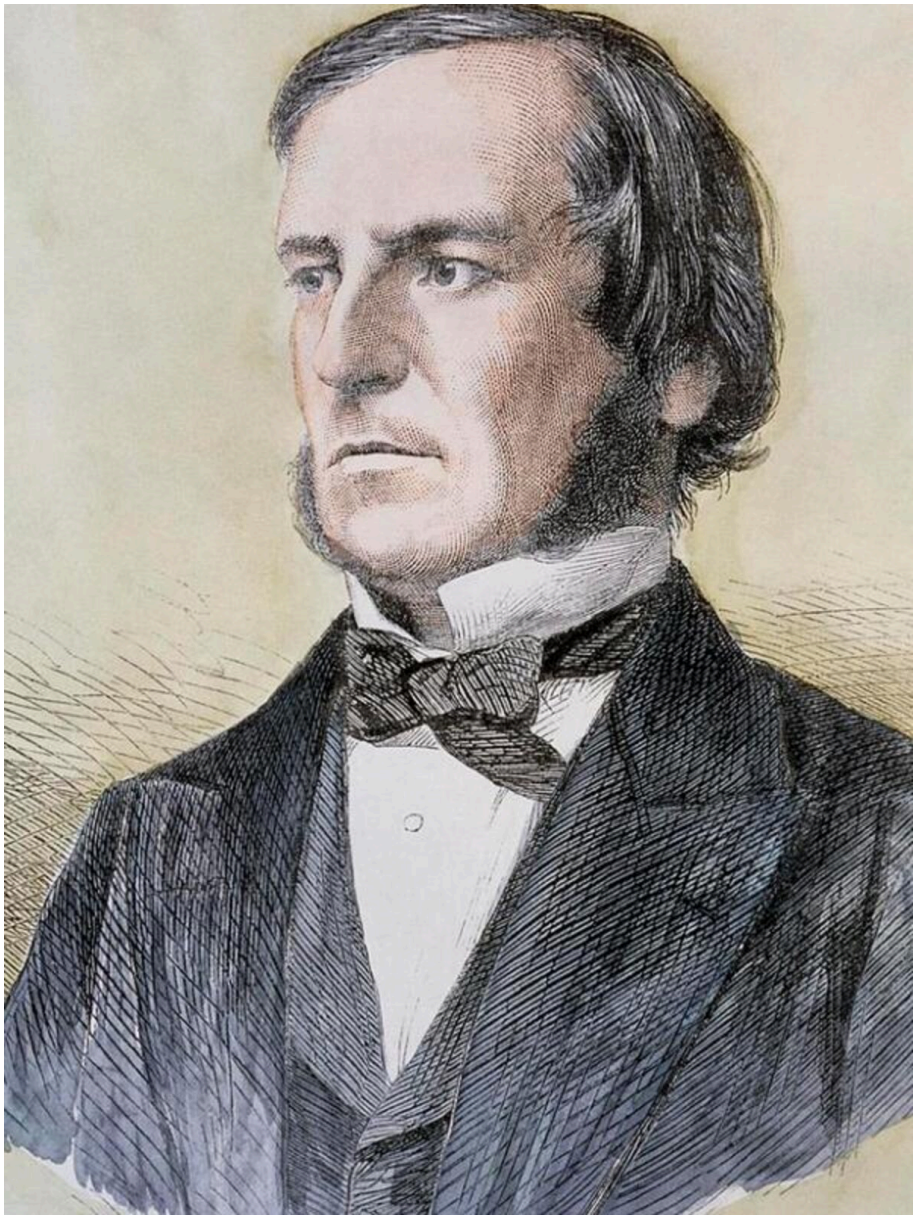

Figure C6.2: George Boole, Originator of Boolean Algebra

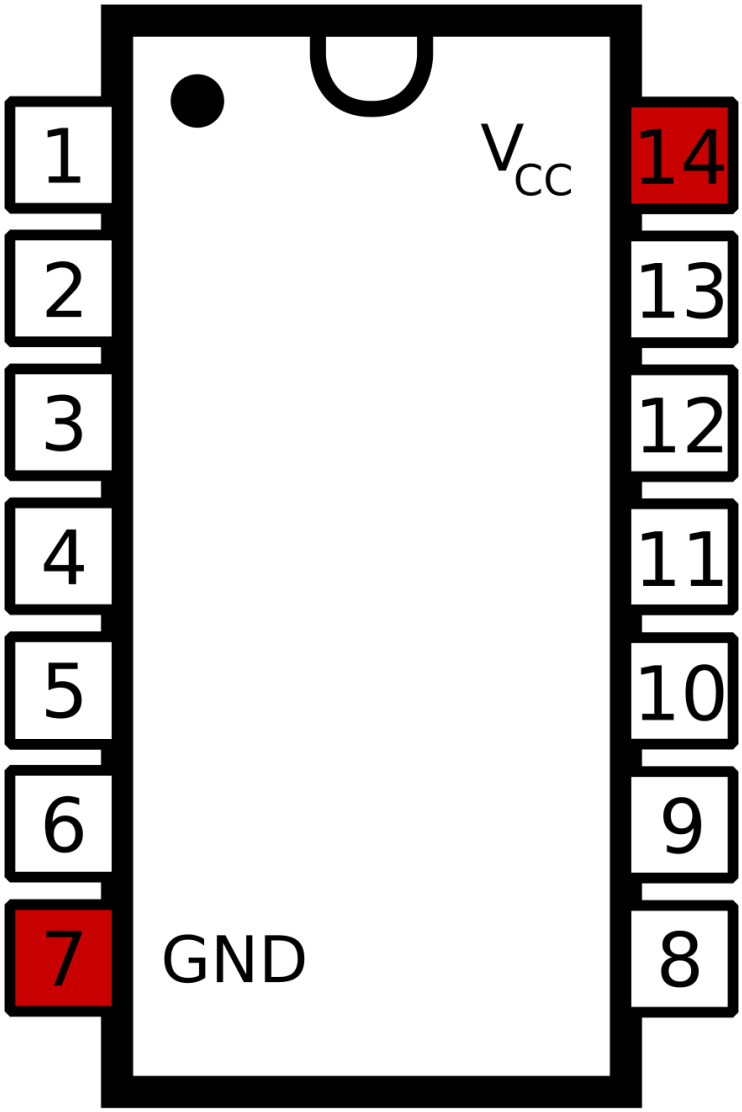

Figure C6.3: Generic 14-Pin TTL Logic Gate IC Pinout

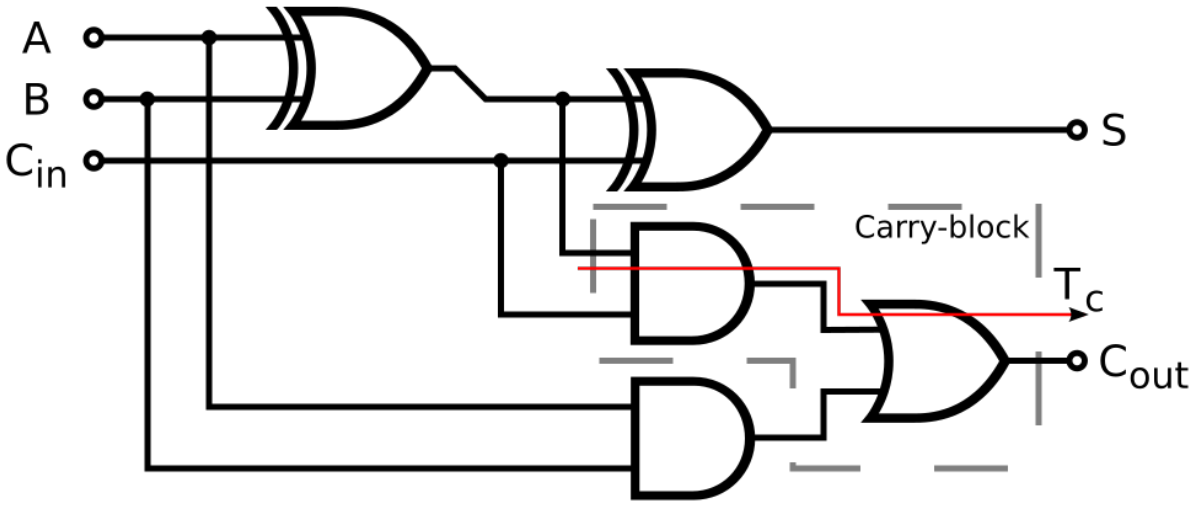

Figure C6.4: Full-Adder Combinational Logic Diagram

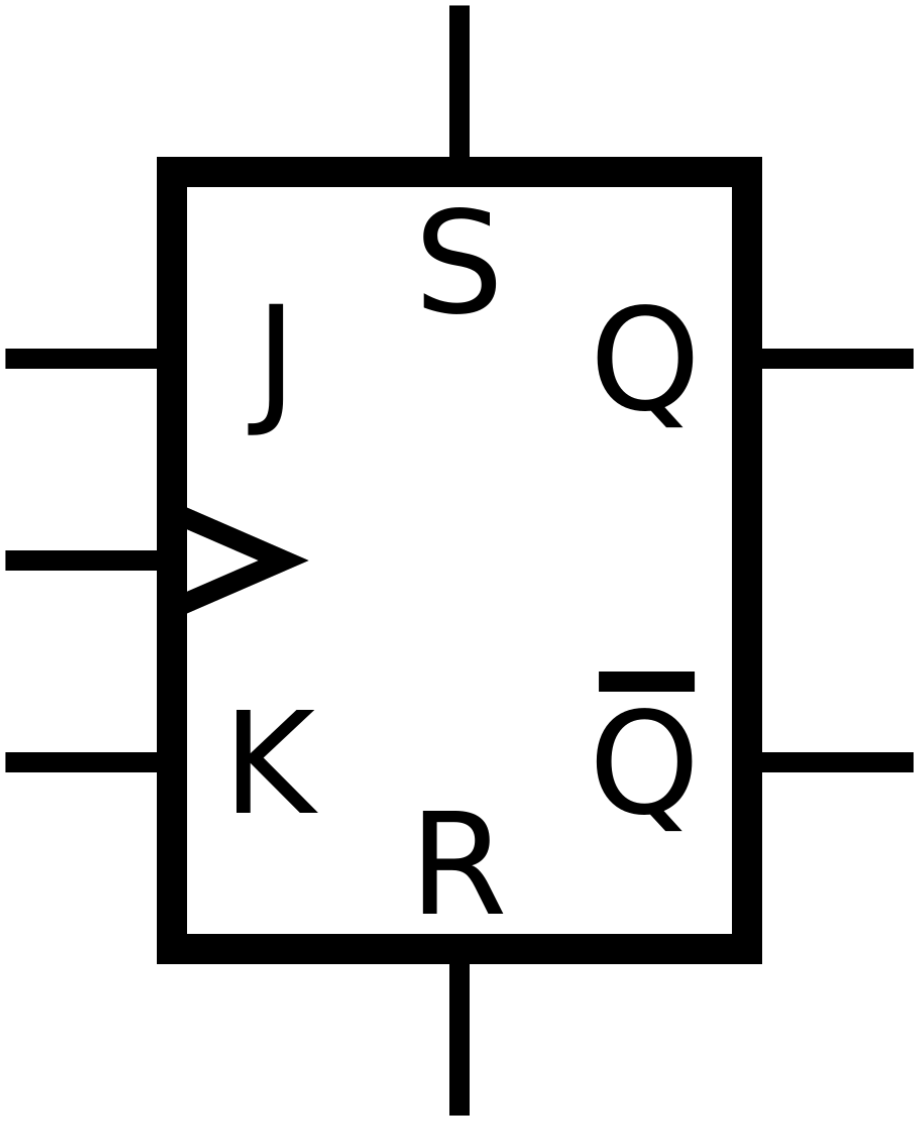

Figure C6.5: JK Flip-Flop Sequential Logic Diagram

Chapter 7: Circuit Analysis

Figure C7.1: Benjamin Franklin Drawing Electricity from the Sky (Benjamin West, c. 1816)

Figure C7.3: Georg Simon Ohm, Namesake of Ohm's Law

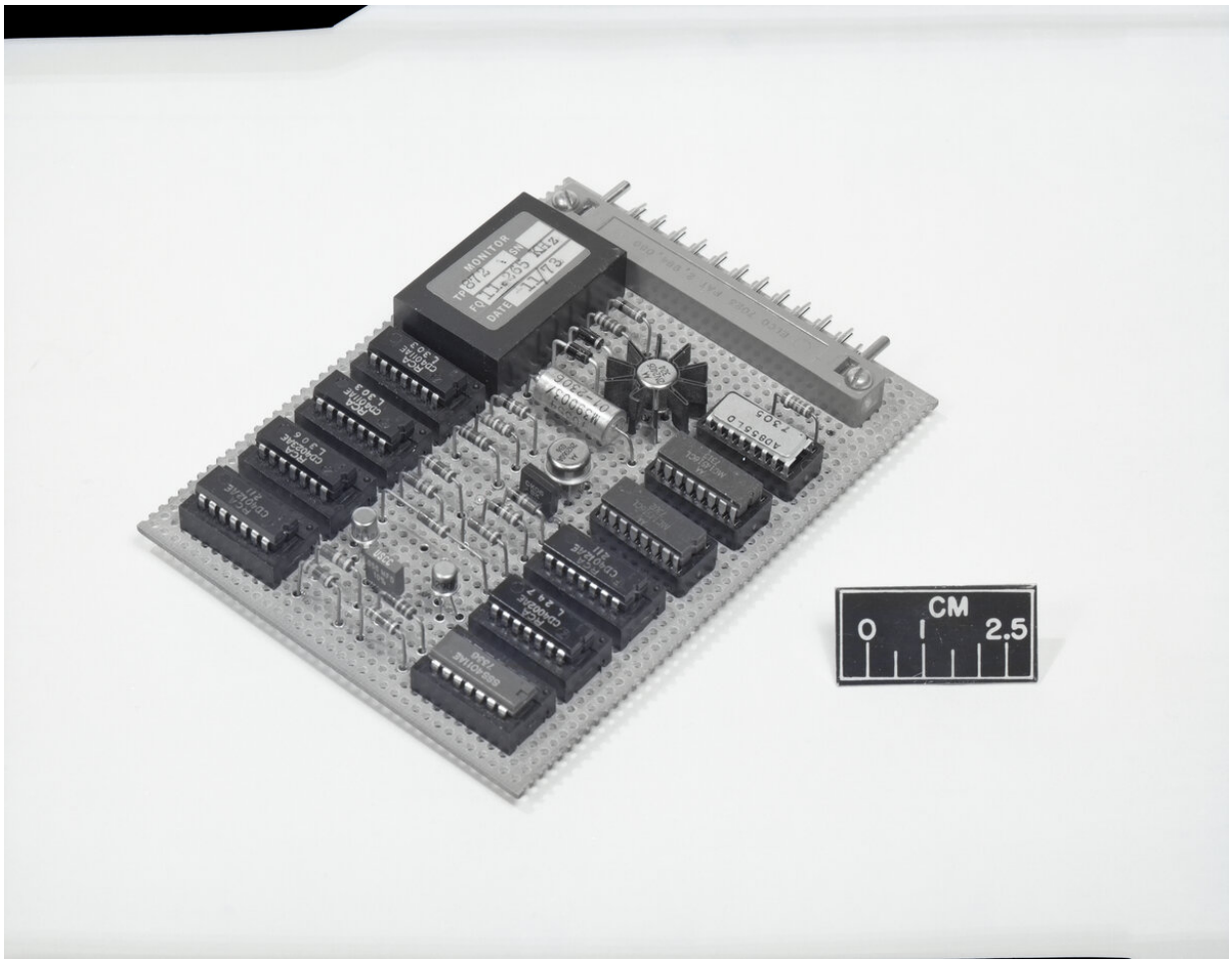

Figure C7.4: Populated DC Circuit Board on Breadboard-Style Perfboard

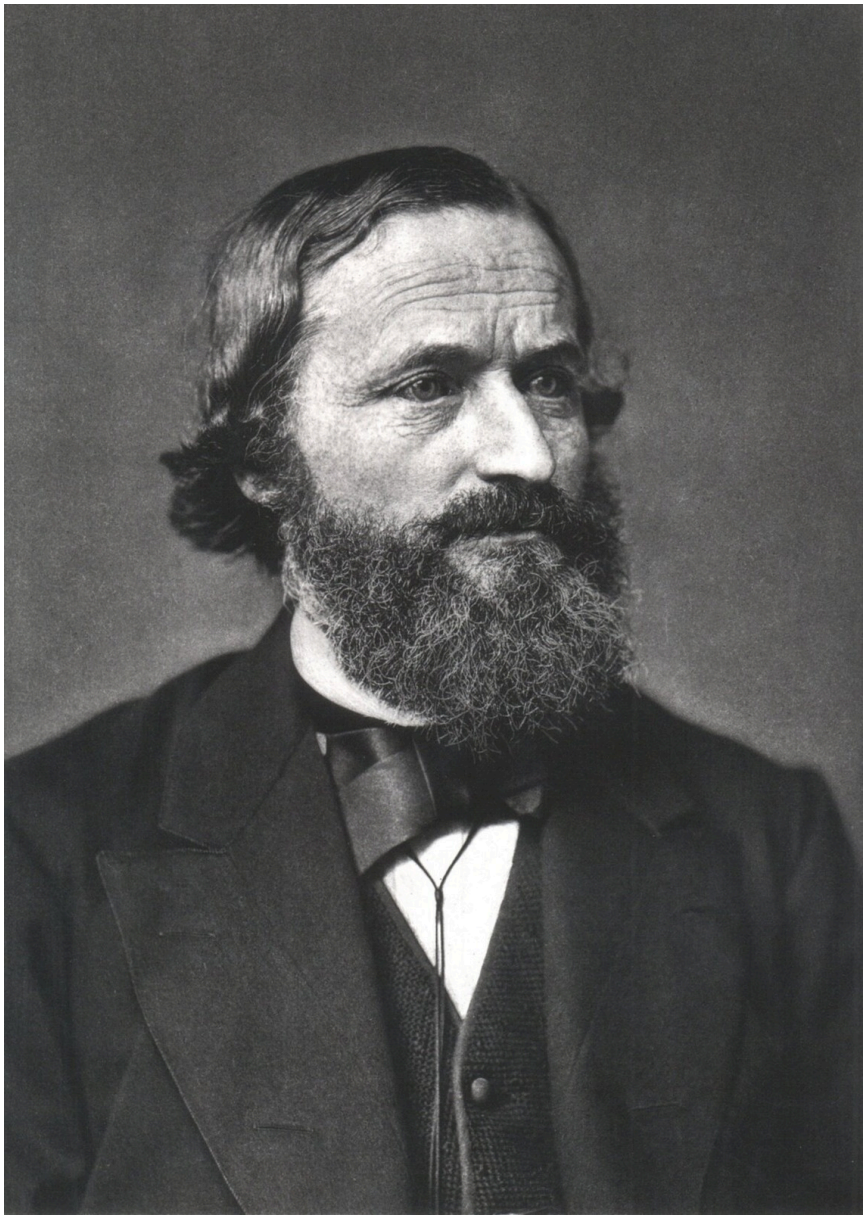

Figure C7.5: Gustav Robert Kirchhoff, Originator of Kirchhoff's Circuit Laws

Figure C7.6: Léon Charles Thévenin, Namesake of Thévenin's Theorem

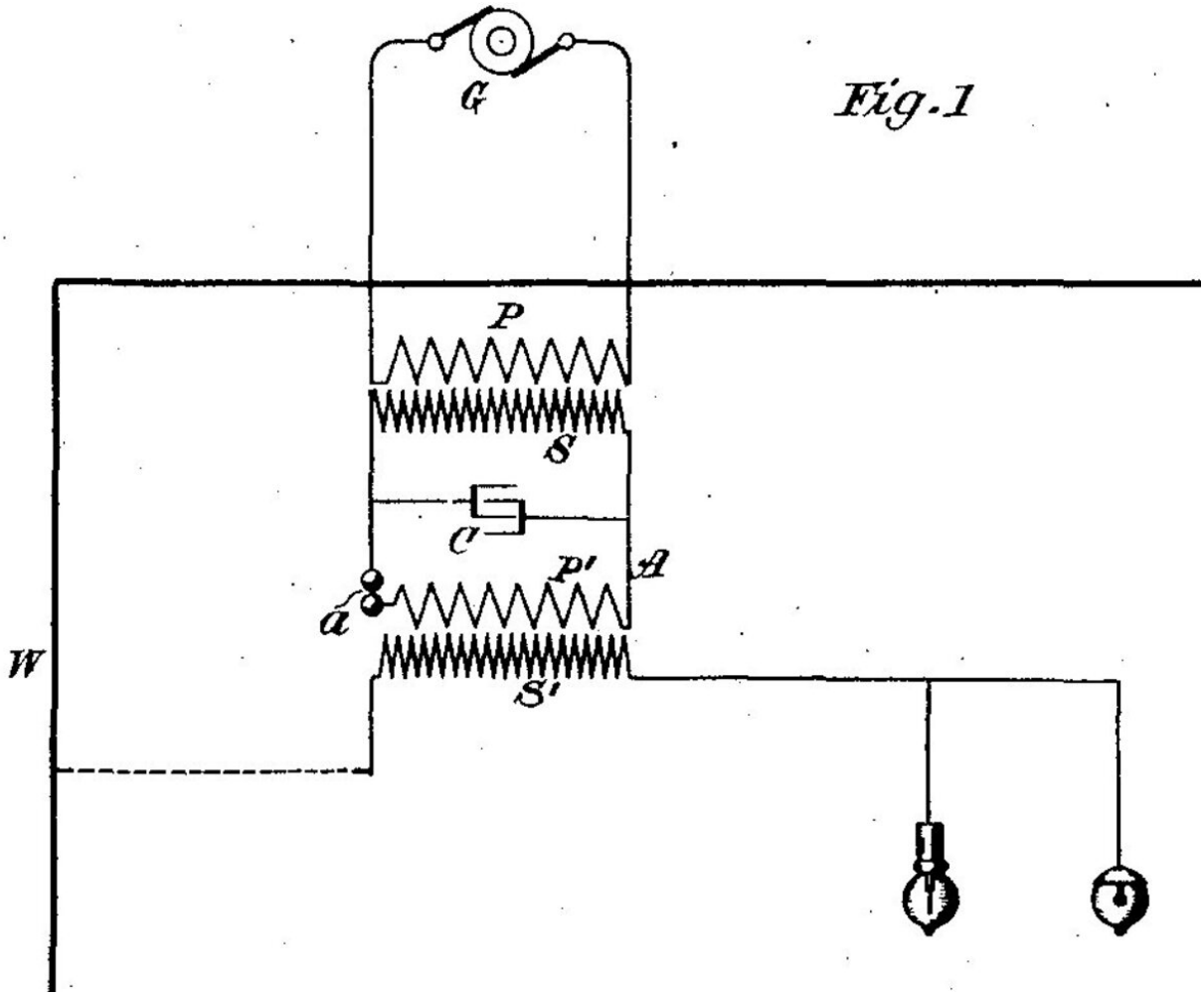

Figure C7.7: Nikola Tesla's 1891 Patent Drawing for a System of Electric (AC) Lighting

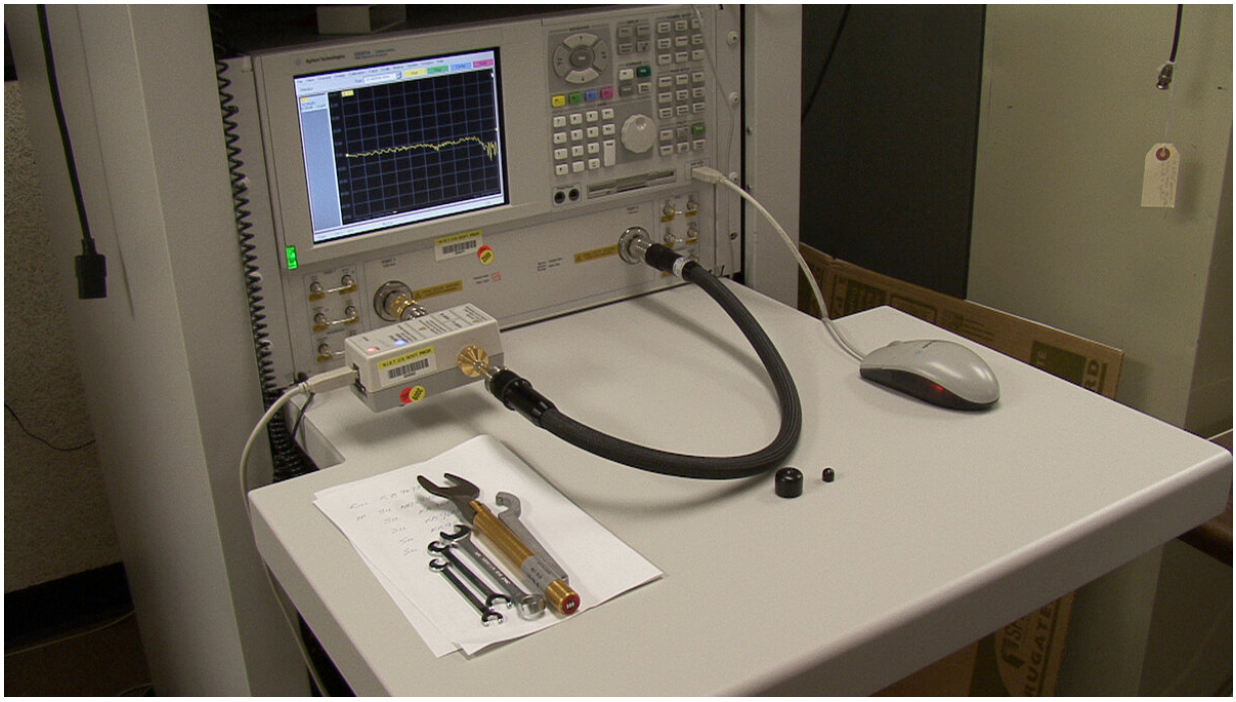

Figure C7.9: Vector Network Analyzer Used for Two-Port Network Calibration

Chapter 8: Signal Processing

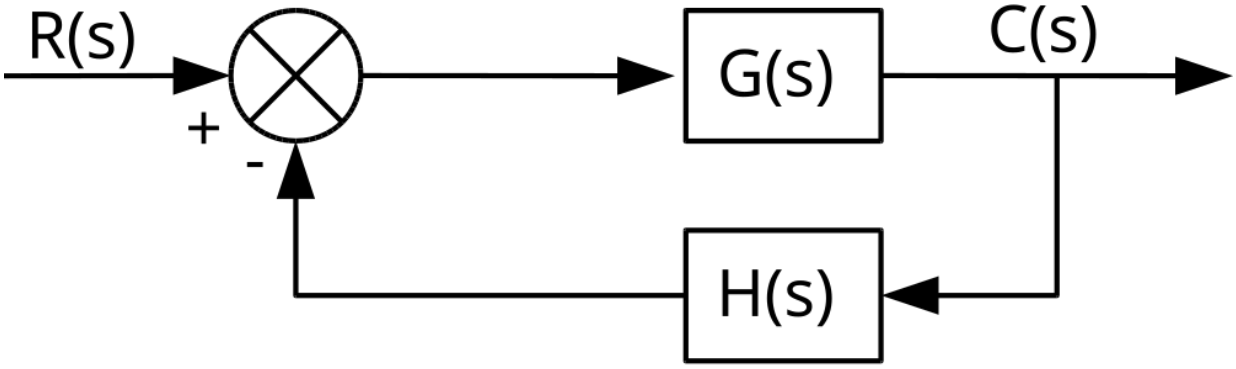

Figure C8.1: General Closed-Loop Feedback System Block Diagram

Figure C8.2: Joseph Fourier, Originator of Fourier Analysis

Figure C8.3: Pierre-Simon Laplace, Namesake of the Laplace Transform

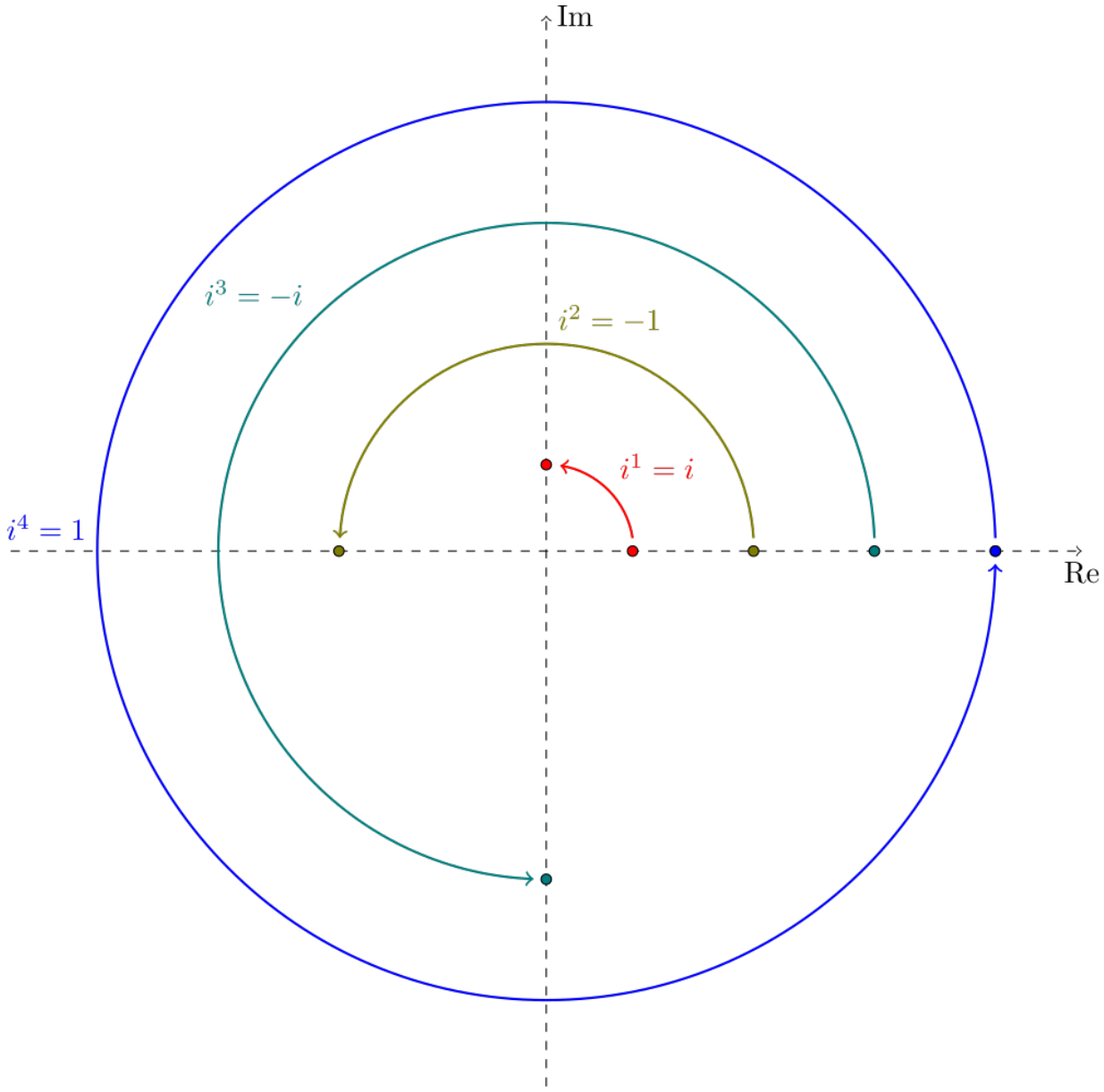

Figure C8.4: Rotations on the Complex Plane — Basis for the Z-Plane Used in the Z-Transform

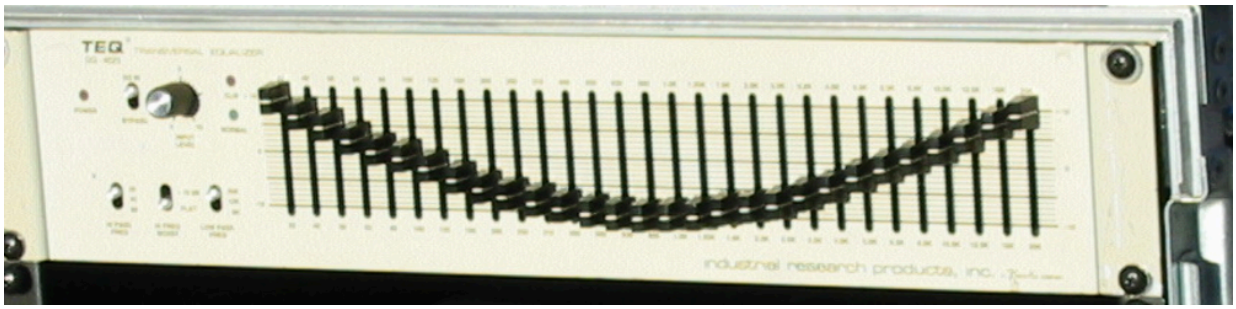

Figure C8.5: Graphic Equalizer — a Bank of Adjustable Digital/Analog Filters

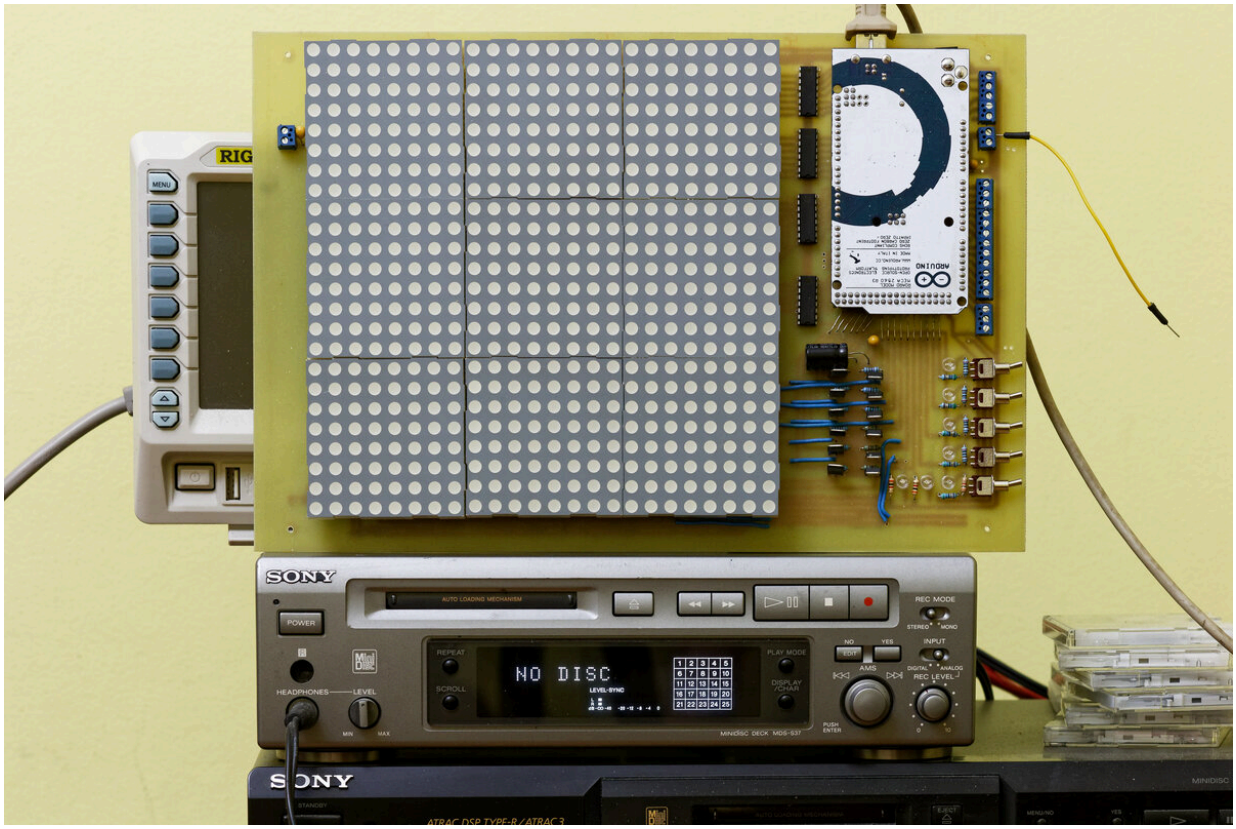

Figure C8.6: DIY Arduino-Based LED Audio Spectrum Analyzer

Chapter 9: Electromagnetics

Figure C9.1: Van de Graaff Generator — Electrostatic Charge Demonstration

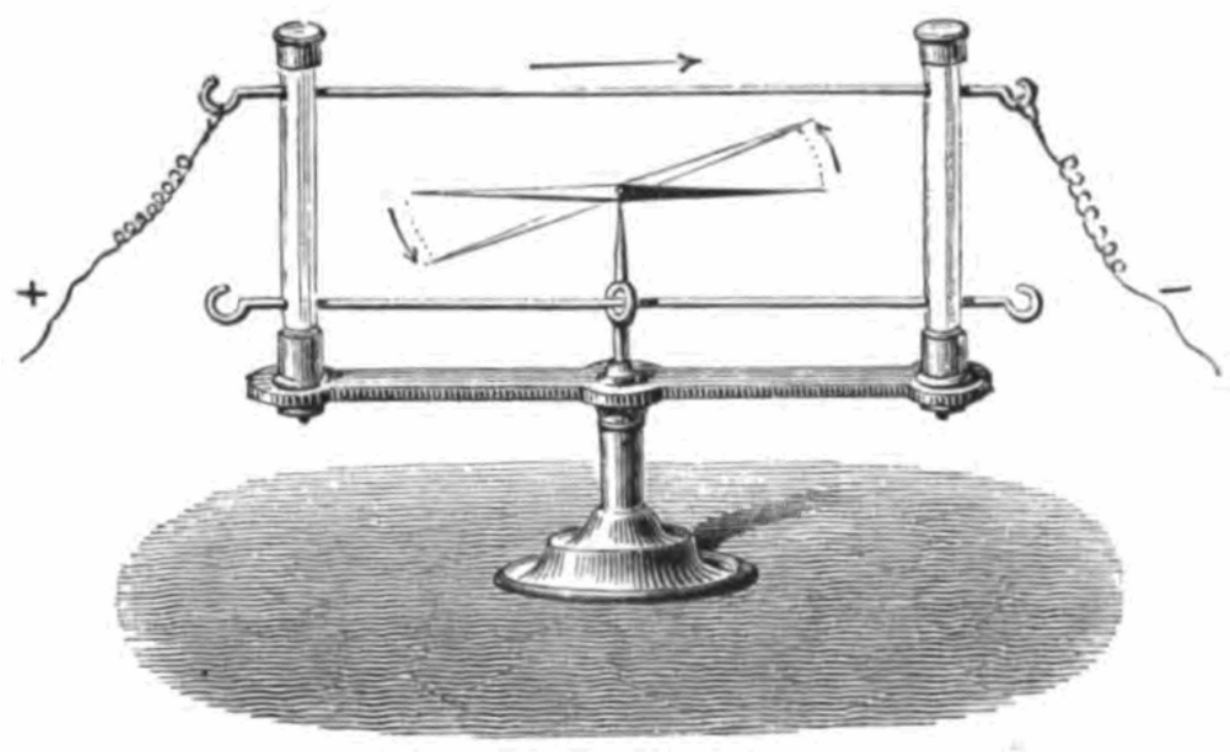

Figure C9.2: Ørsted's Experiment — Current-Carrying Wire Deflecting a Compass Needle

Figure C9.3: James Clerk Maxwell, Formulator of Maxwell's Equations

Figure C9.4: Heinrich Hertz, First to Demonstrate Electromagnetic Waves

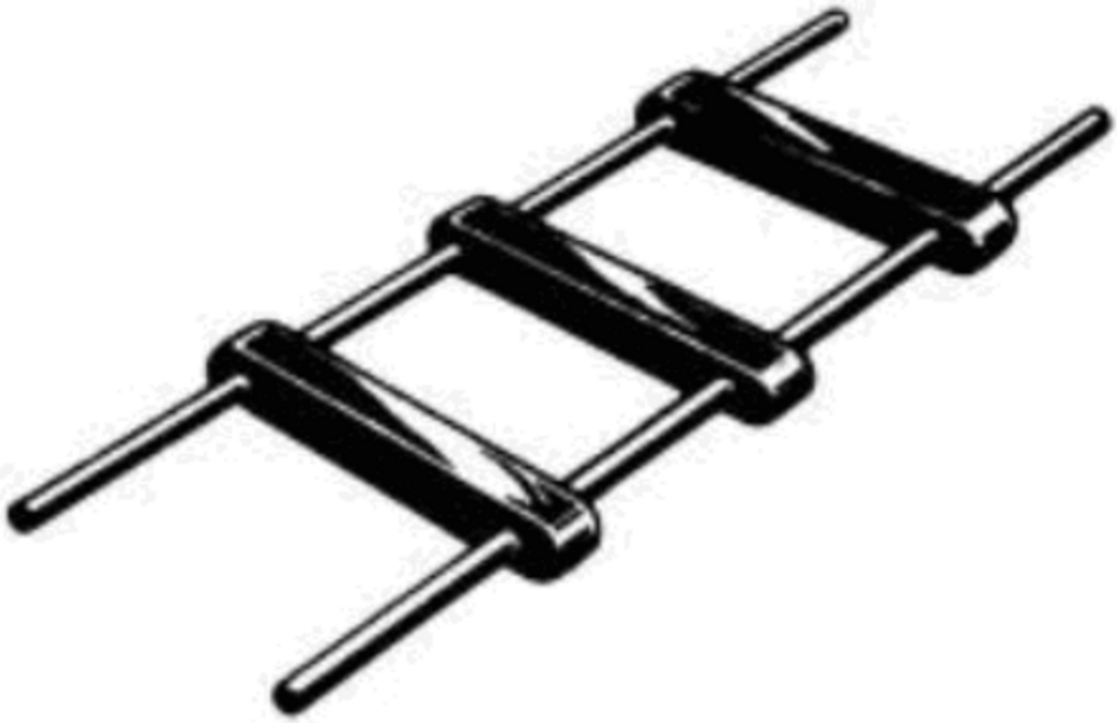

Figure C9.5: Ladder Line — an Open-Wire Balanced Transmission Line

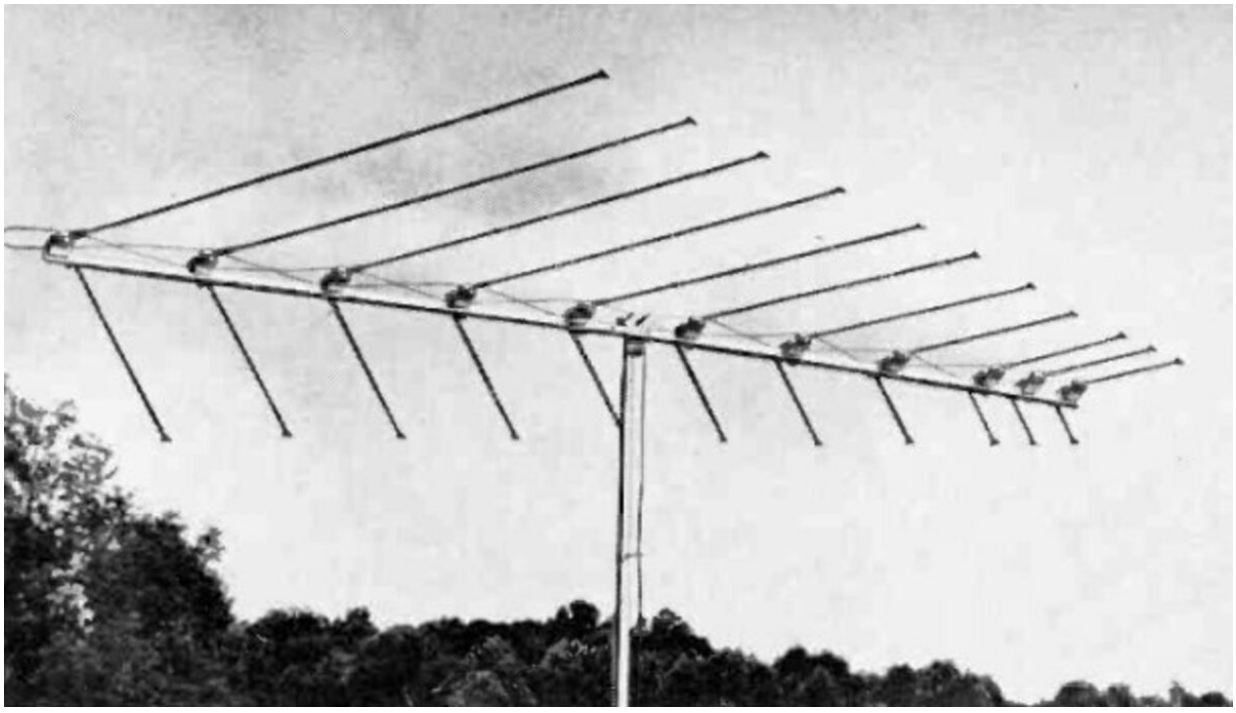

Figure C9.6: Log-Periodic VHF TV Antenna (1963)

Chapter 10: Power Electronics

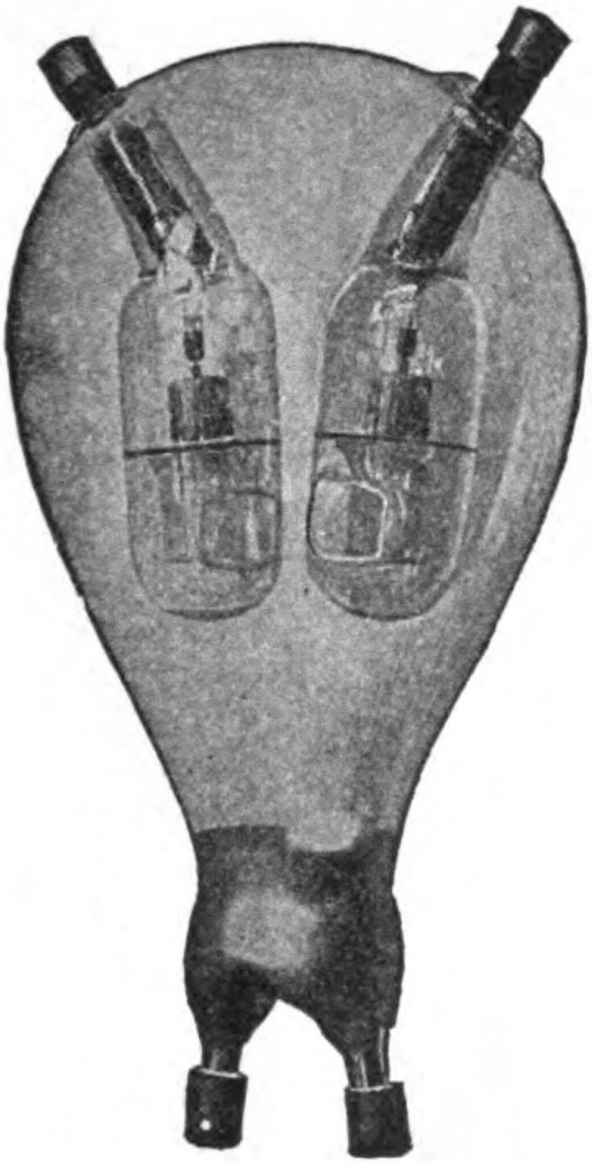

Figure C10.2: Twin-Anode Mercury-Arc Rectifier Bulb

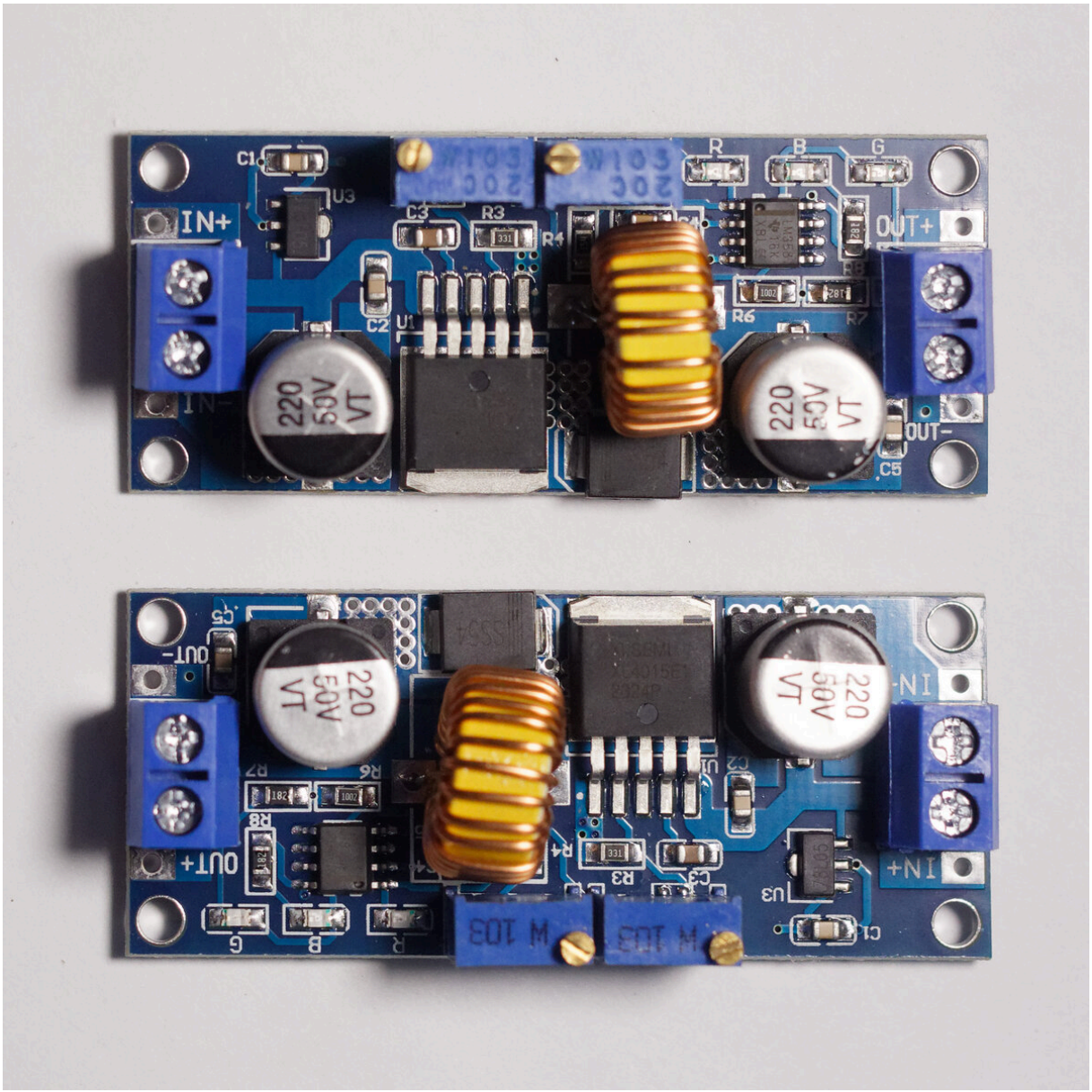

Figure C10.3: XL4015-Based CC/CV Buck Converter Modules

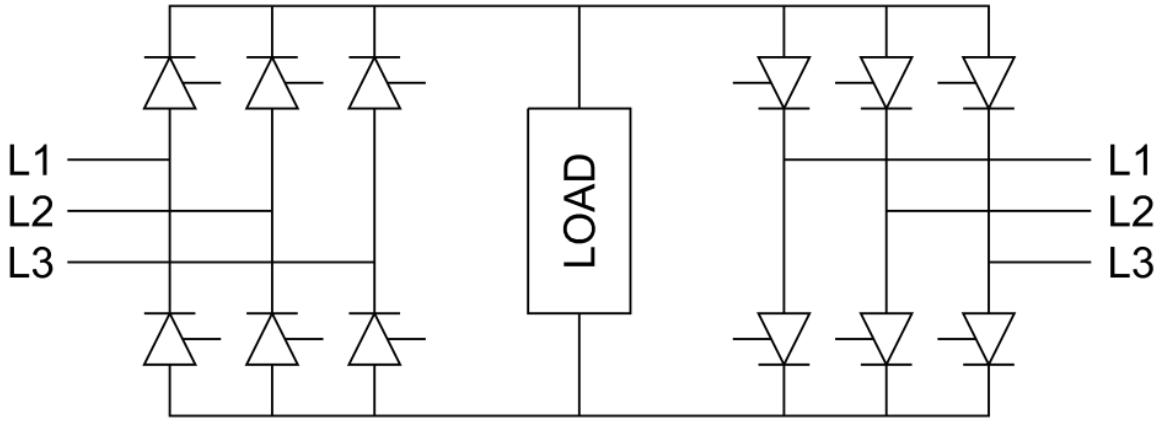

Figure C10.5: Cycloconverter Circuit Diagram (Direct AC-AC Conversion)

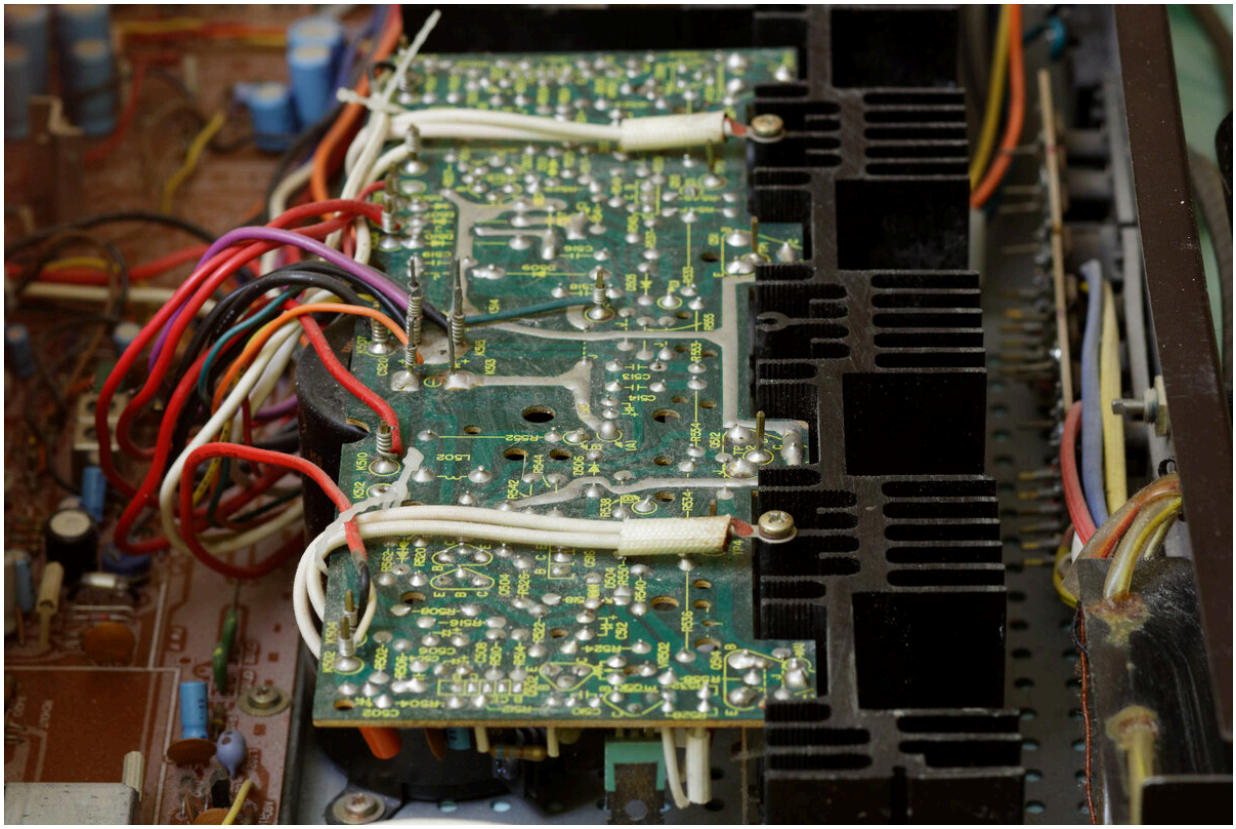

Figure C10.6: Heatsinked Power Amplifier Output Transistors — Thermal Management in Power Electronics

Figure C10.7: Pole-Mounted Power Factor Correction Capacitor Bank

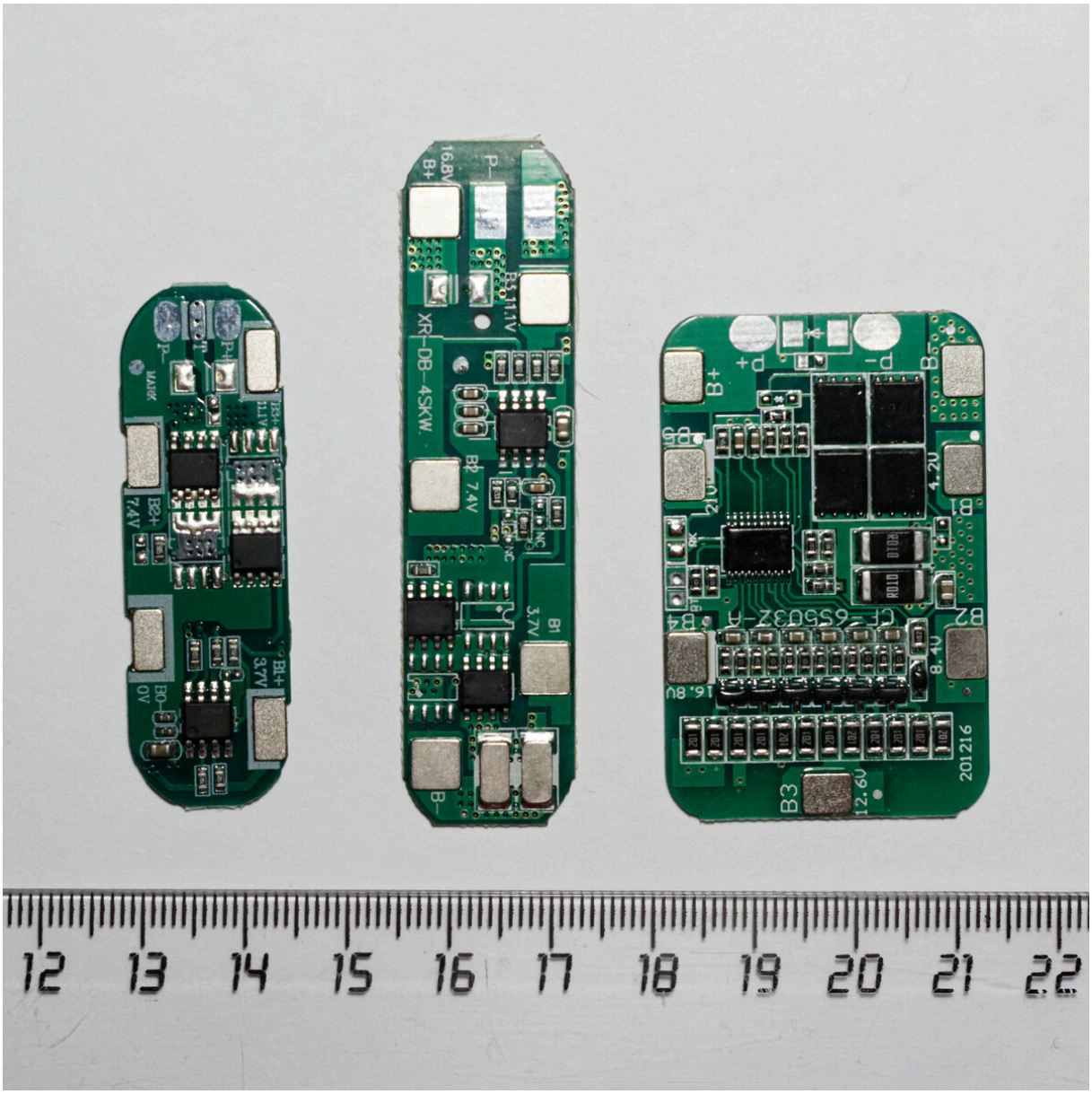

Figure C10.8: Generic Lithium Battery Management System (BMS) Boards, 3S/4S/6S

Figure C10.10: Tesla V3 Supercharger — High-Power DC Fast Charging Station

Figure C10.11: Commercial Supercapacitor (Ultracapacitor) Products

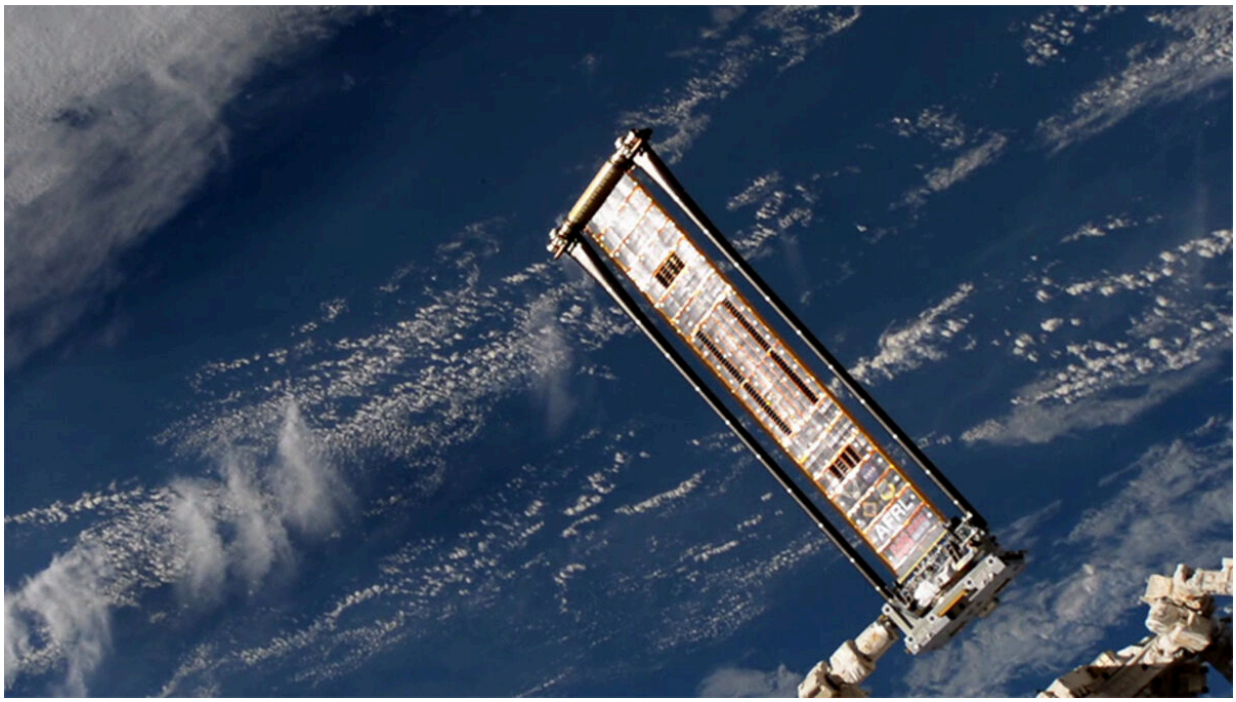

Figure C10.12: Roll-Out Solar Array (ROSA) Deployed on the International Space Station

Chapter 11: Instrumentation and Measurement

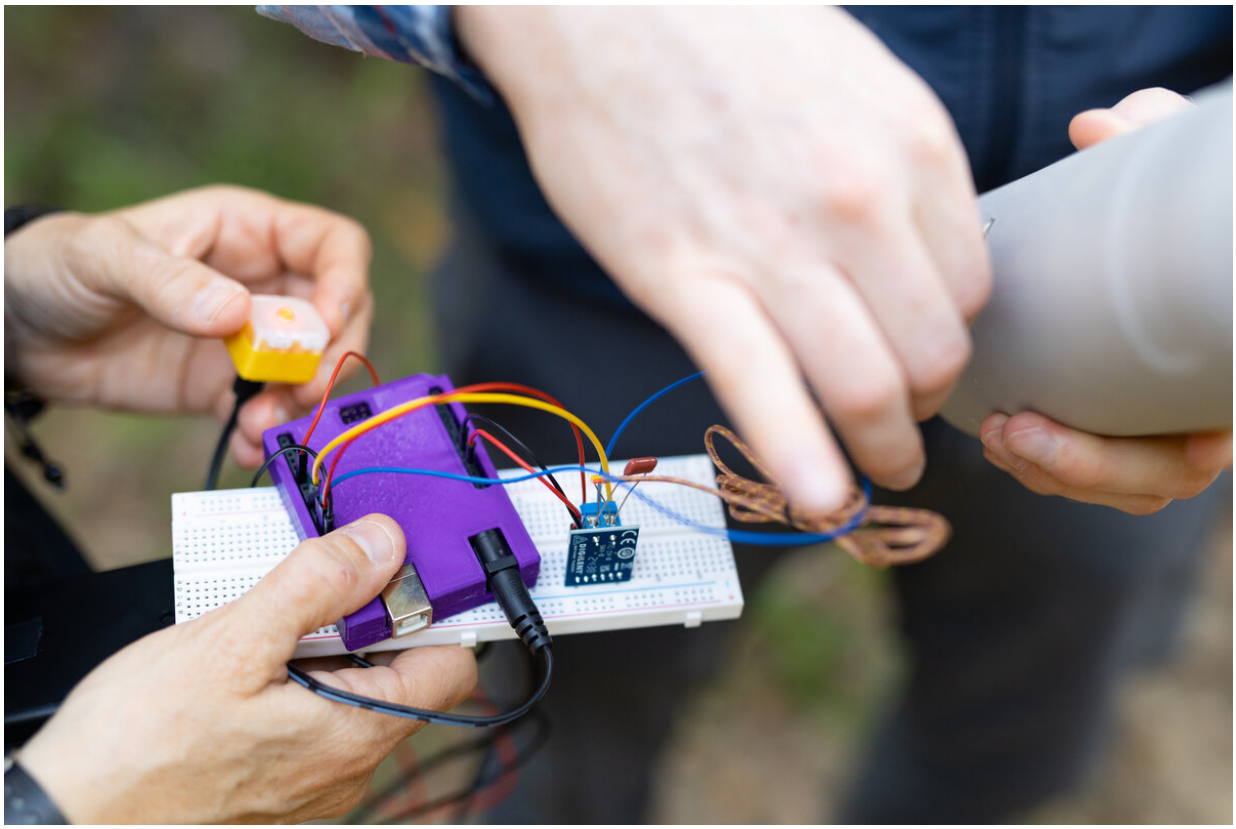

Figure C11.2: Breadboarded Sensor Module with Digital Readout — Field Data Collection

Figure C11.4: Handheld Digital Multimeter — a Common Measurement Instrument

Figure C11.5: Wind-Tunnel Data Acquisition System Computer Room

Chapter 12: Electric Motors

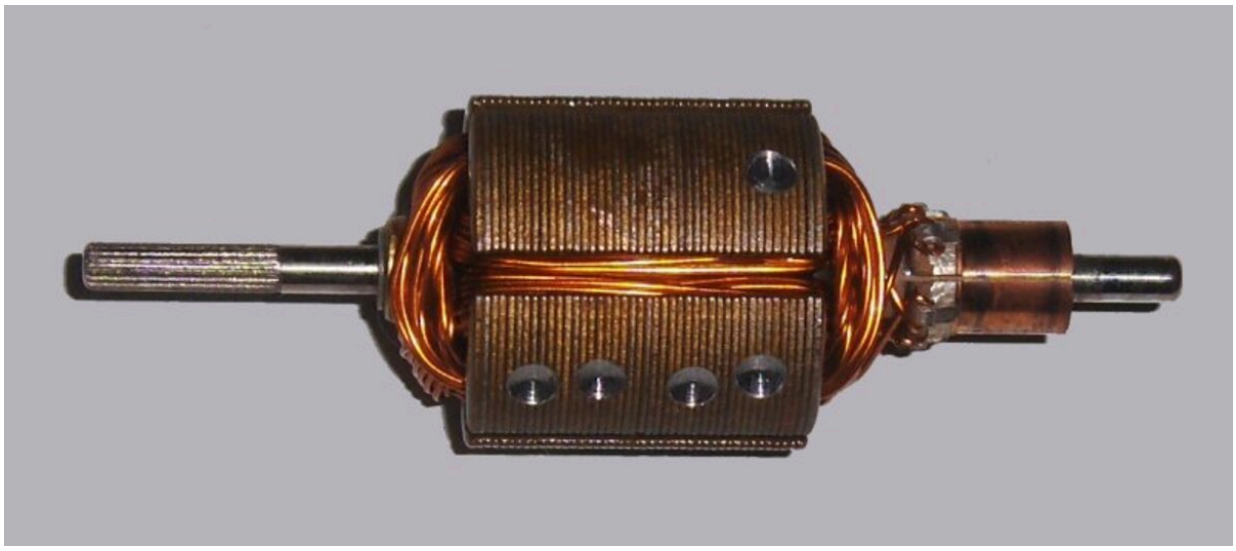

Figure C12.1: DC Motor Armature with Commutator and Windings

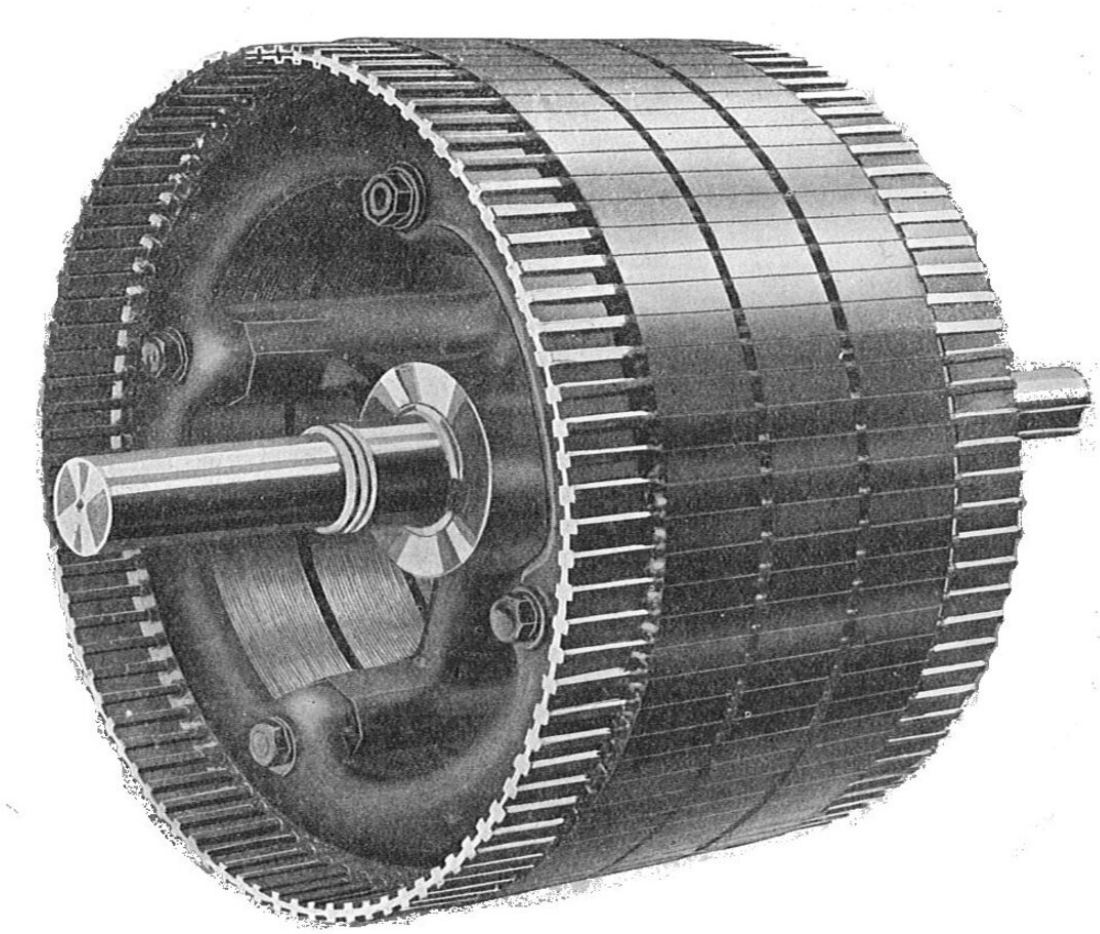

Figure C12.2: Squirrel-Cage Induction Motor Rotor (1909)

Chapter 13: Operational Amplifiers

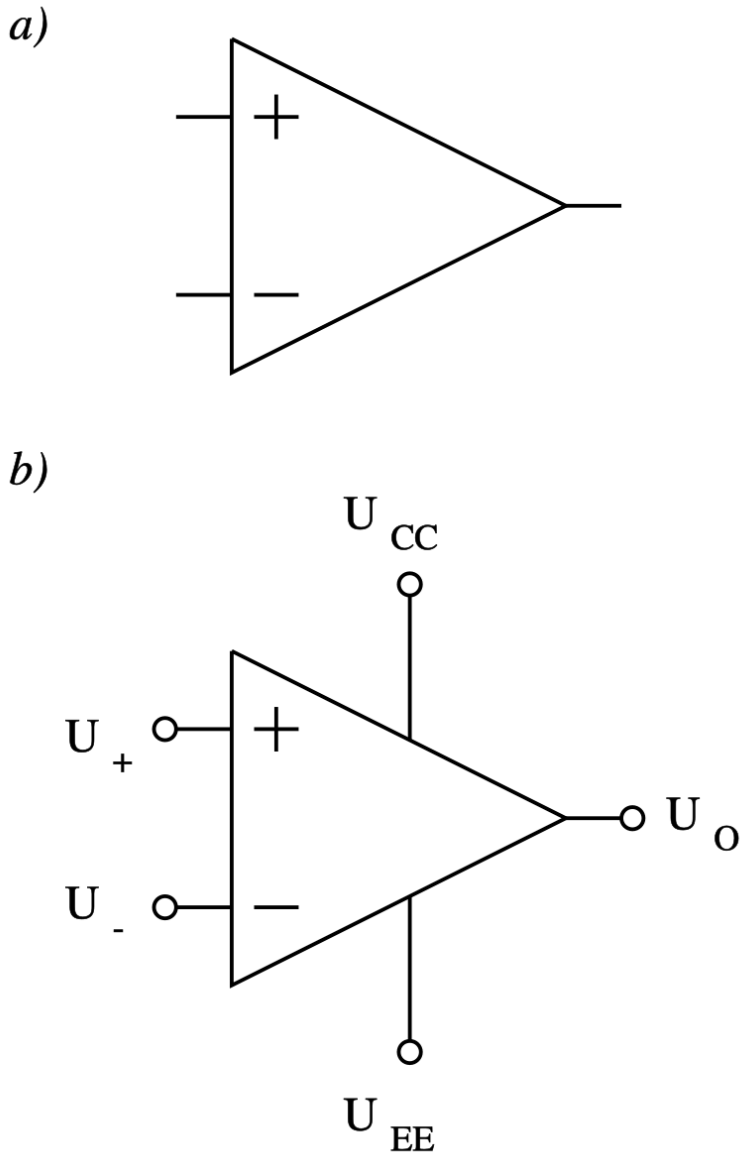

Figure C13.1: Ideal Op-Amp Symbol, With and Without Supply Rails Shown

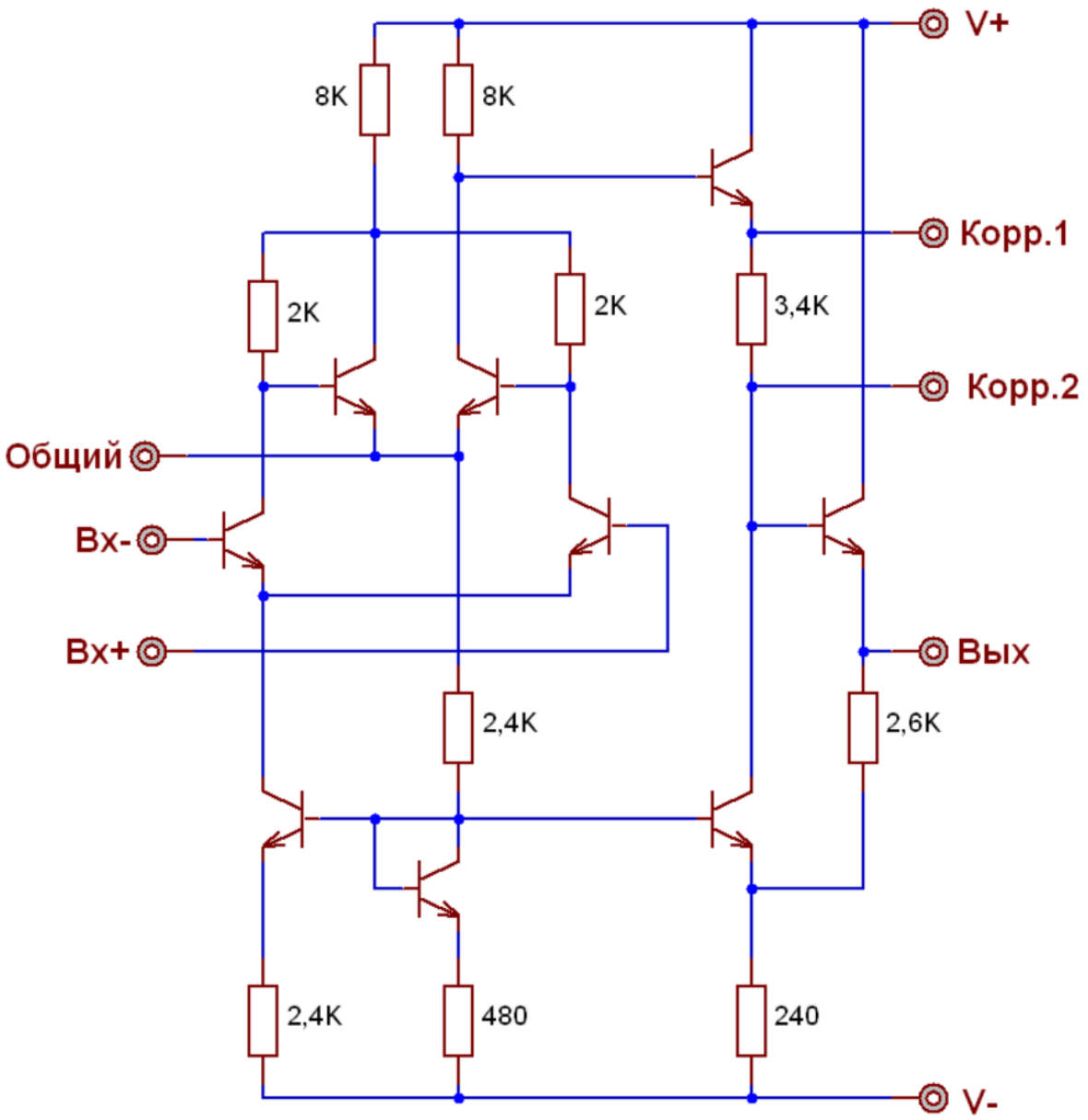

Figure C13.2: Internal Transistor-Level Schematic of the μA702, the First Monolithic Op-Amp (Bob Widlar, 1964)

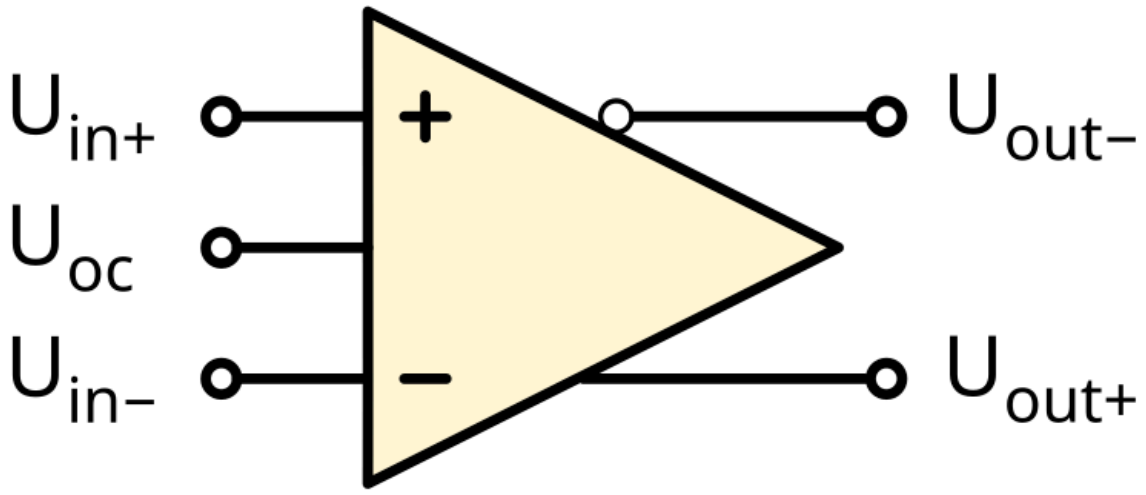

Figure C13.4: Fully-Differential Amplifier Symbol

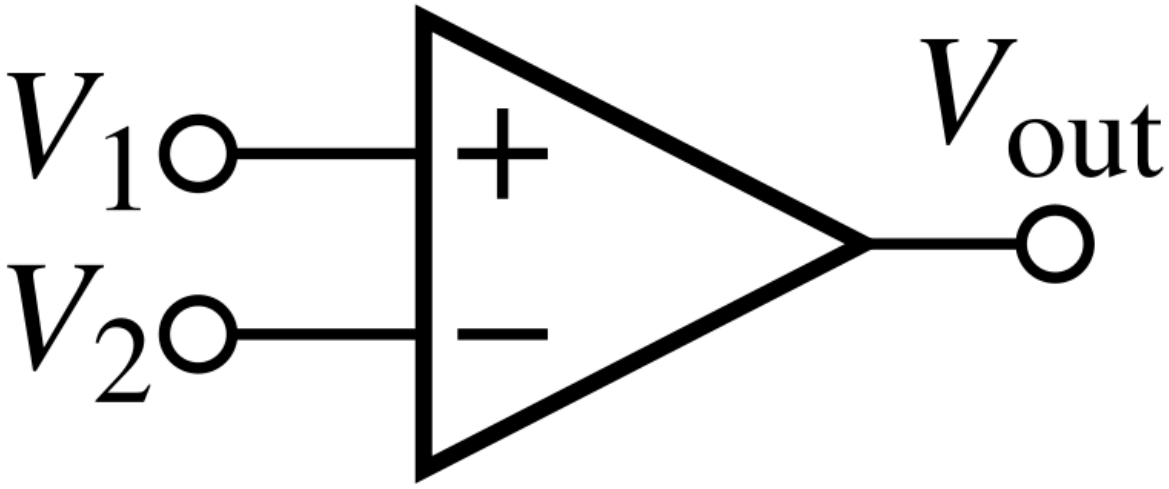

Figure C13.6: Op-Amp Used as a Comparator — Symbol and Convention

Chapter 14: National Electrical Code (NEC)

Figure C14.2: Weatherproof Electrical Device Box for Conductor Terminations

Figure C14.6: Pad-Mounted Distribution Transformer

Chapter 15: Networking

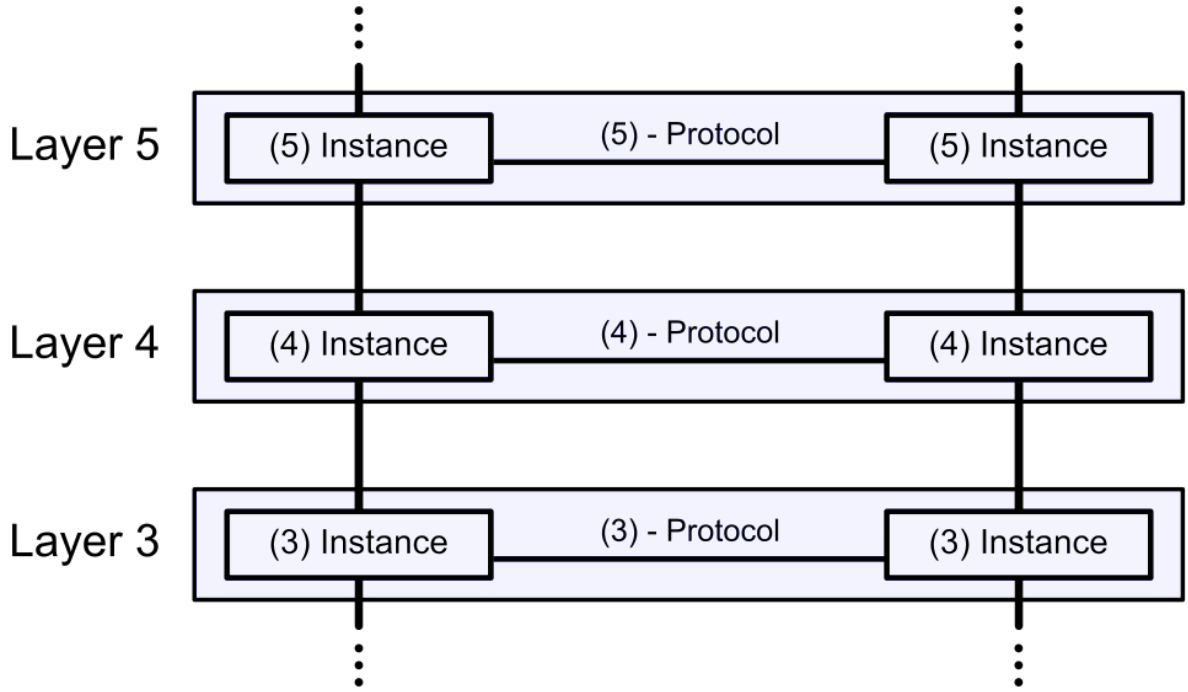

Figure C15.1: Peer-Layer Protocol Communication — the OSI Model's Layered Abstraction

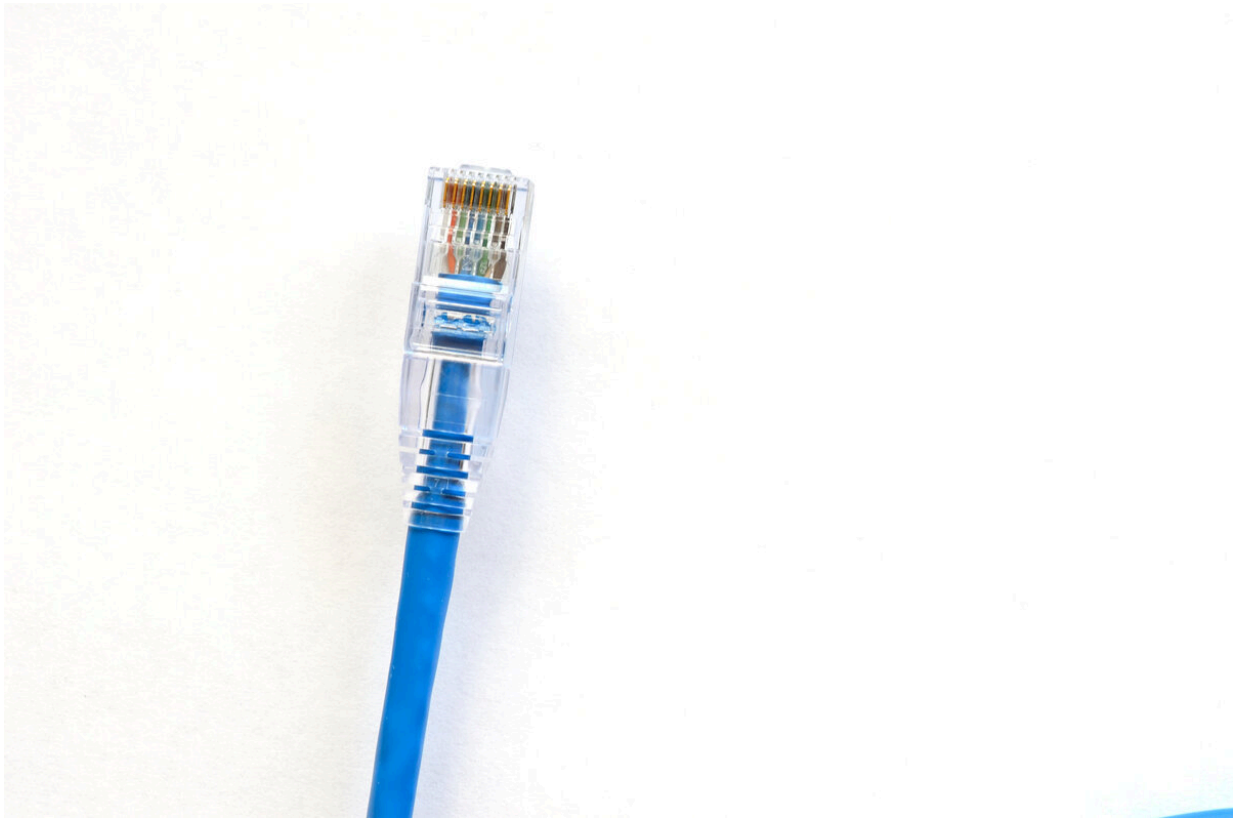

Figure C15.2: Twisted-Pair Copper Ethernet Cable

Figure C15.3: Installing Fiber-Optic Cabling for Rural Broadband (USDA Rural Development, 2024)

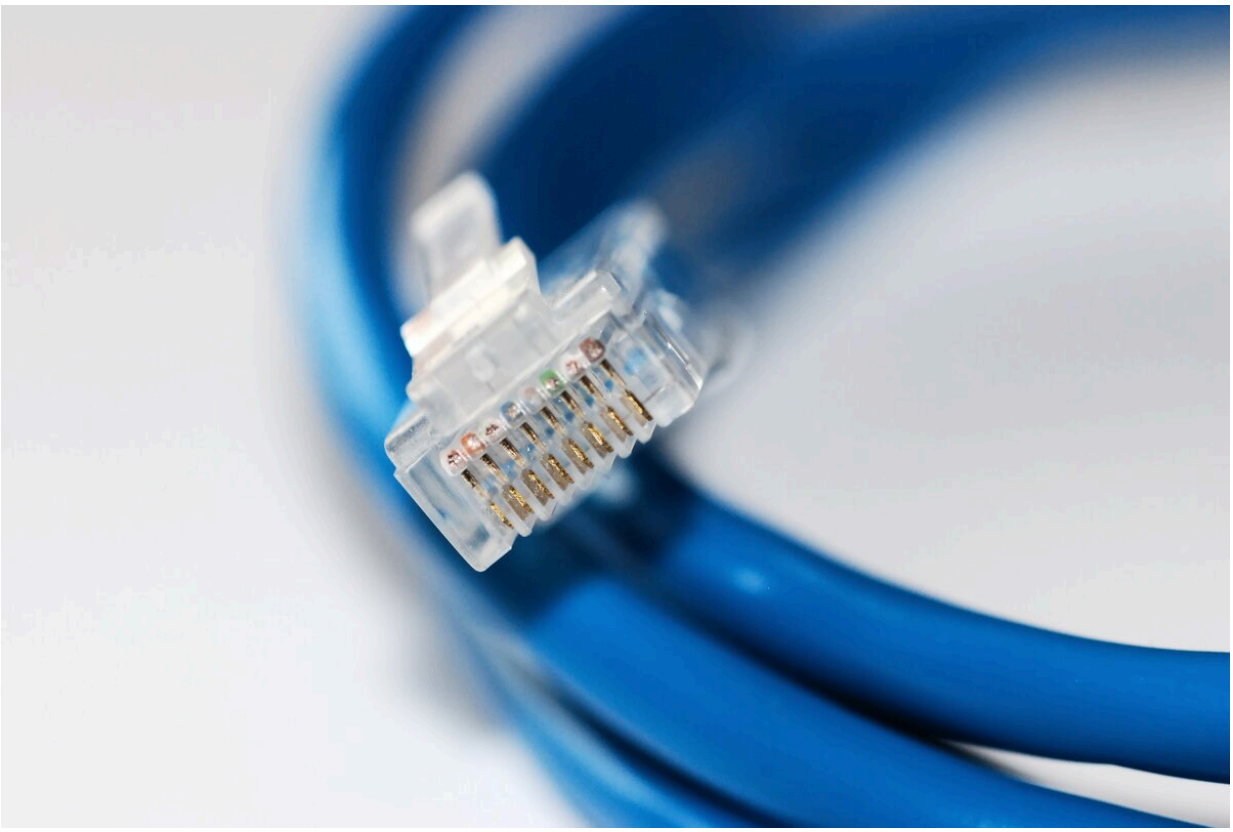

Figure C15.5: RJ45 (8P8C) Connector Used for Ethernet Twisted-Pair Cabling

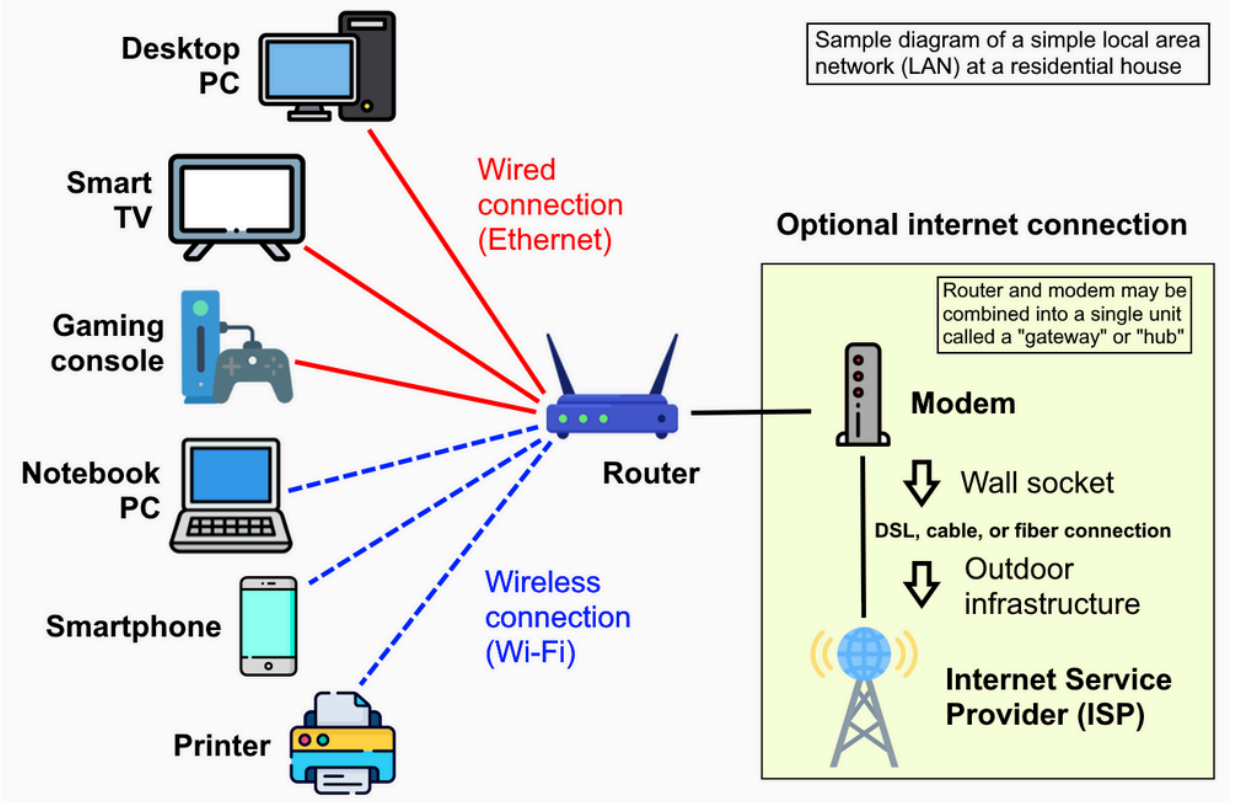

Figure C15.6: Example Home LAN / IP Network Diagram

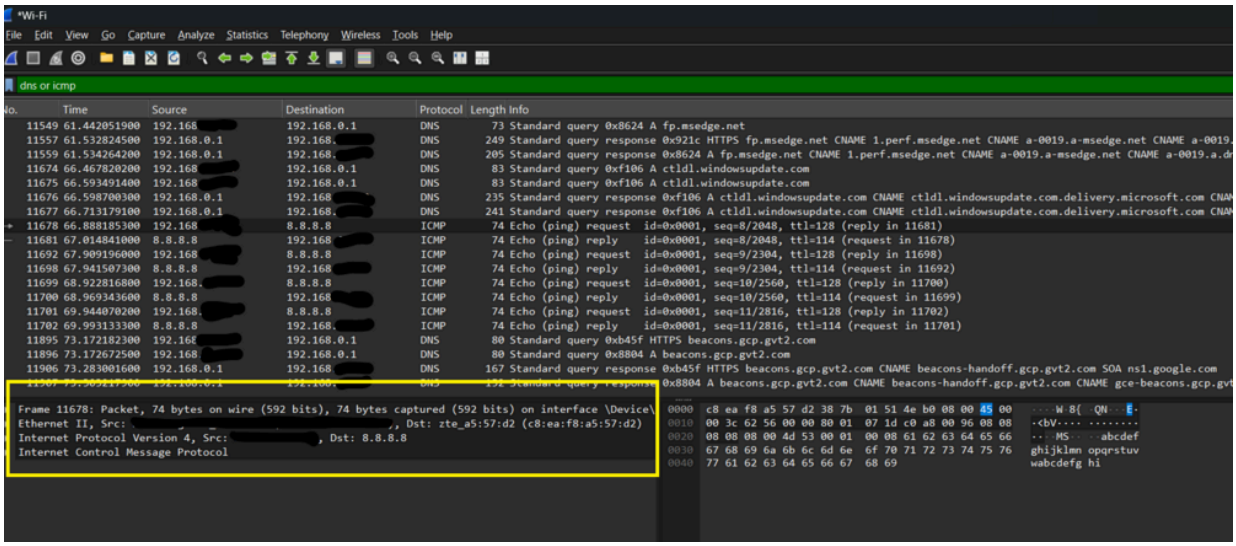

Figure C15.7: Wireshark Packet Capture — Inspecting Transport/Network-Layer Protocol Traffic

Figure C15.8: Ceiling-Mounted Wireless (Wi-Fi) Access Point

Figure C15.10: Network Cable Tester — Verifying Link Performance and Wiring Continuity

Chapter 16: Antenna Design and Principles

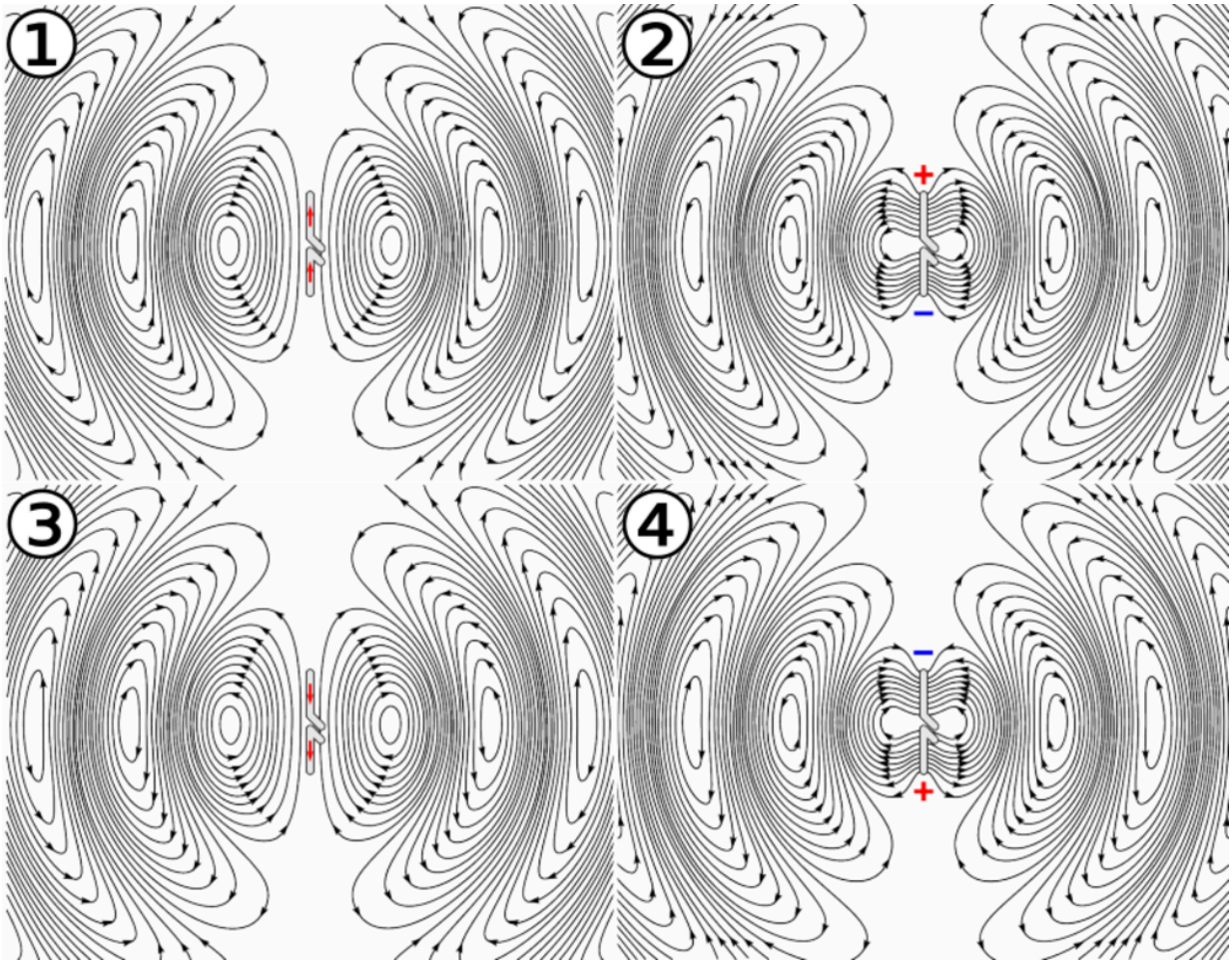

Figure C16.1: Half-Wave Dipole Antenna Field Patterns (Transmit and Receive)

Figure C16.2: UHF Ground-Plane (Quarter-Wave Whip) Antenna

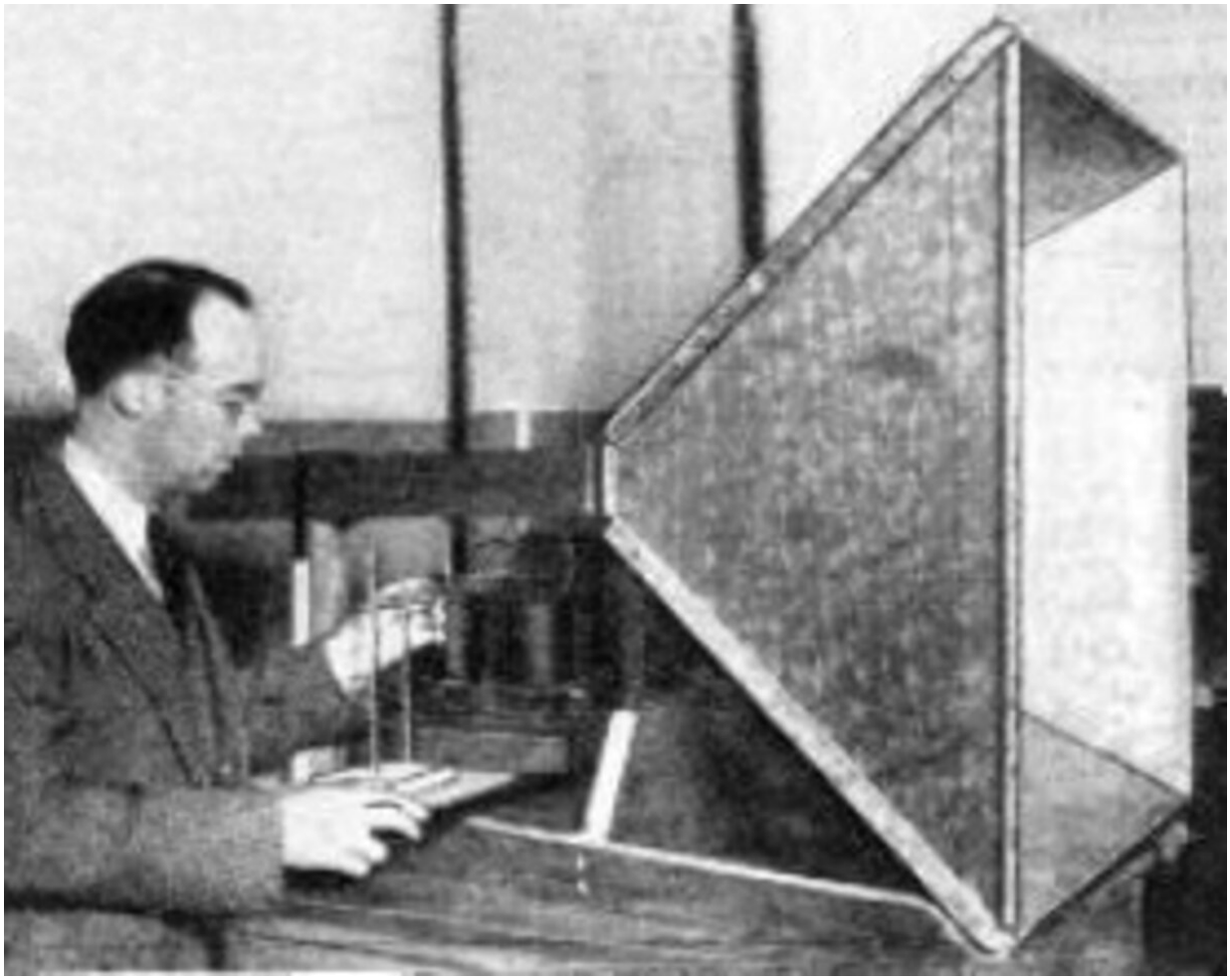

Figure C16.3: Wilmer Barrow and the First Modern Horn Antenna (1938)

Figure C16.5: AN/FPS-85 Phased-Array Radar Antenna Building, Eglin AFB

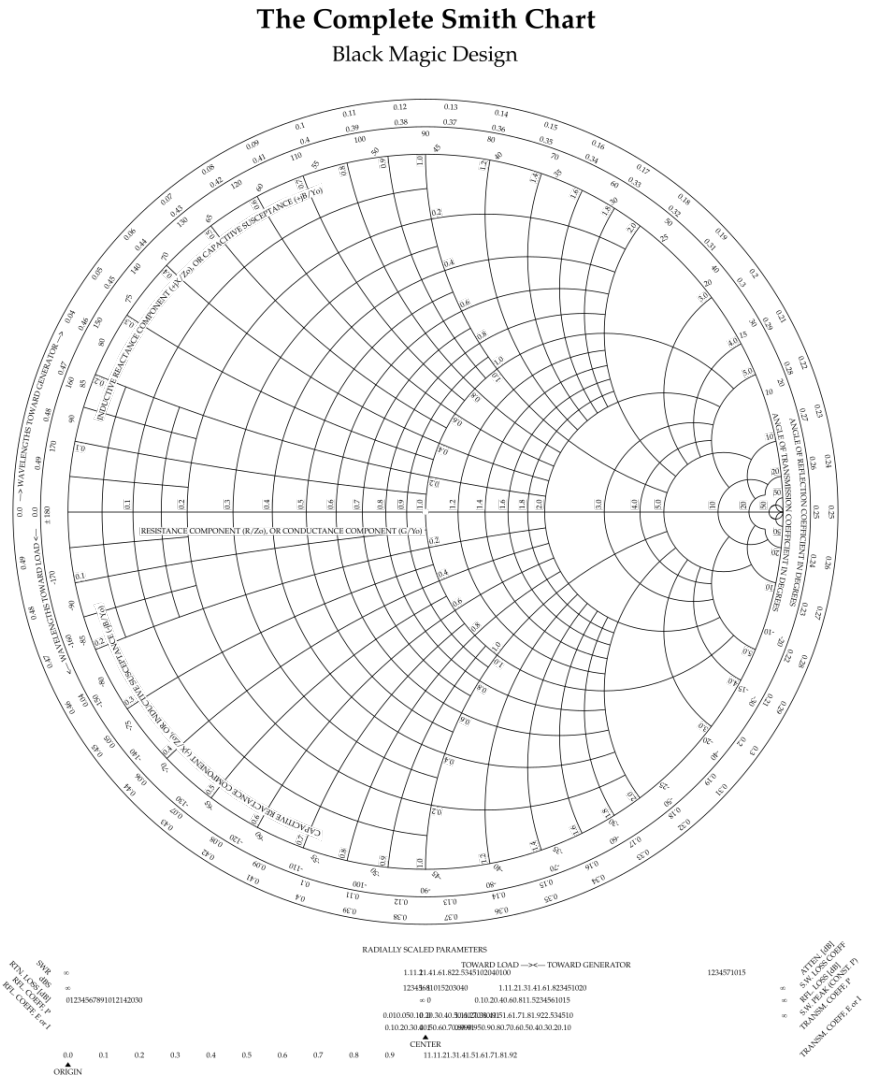

Figure C16.6: Smith Chart Used for Antenna/Transmission-Line Impedance Matching

Chapter 17: Radar Systems

Figure C17.3: PAVE PAWS Radar Operator Console — Processed Track Display

Figure C17.4: WSR-1 Weather Radar and Radiosonde Dome, North Omaha Airport

Figure C17.5: NATO E-3 AWACS — Airborne Early-Warning Radar System

Chapter 18: Optics

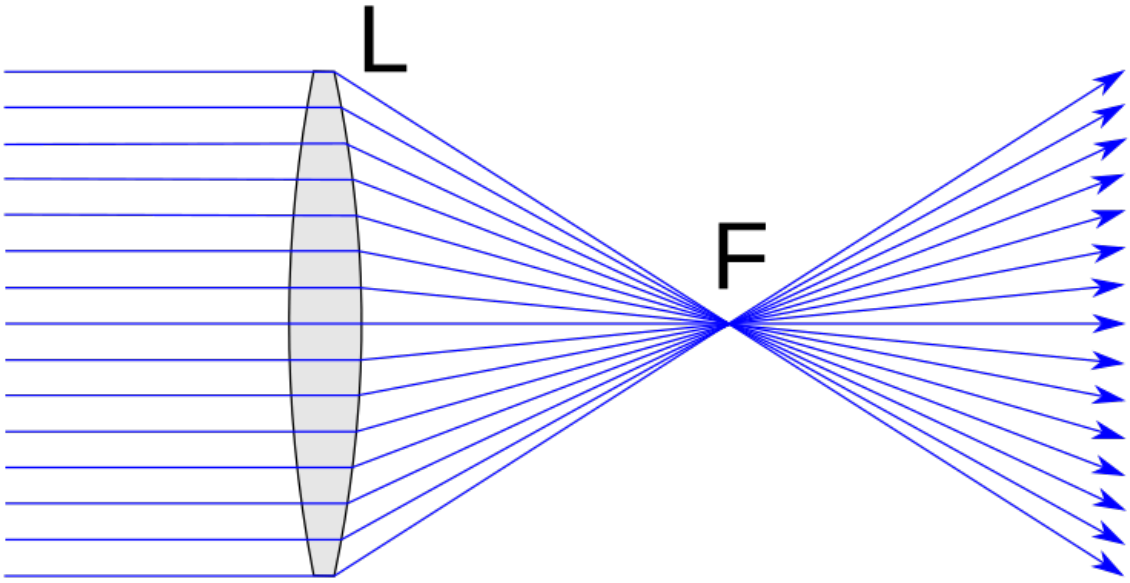

Figure C18.1: Ray Diagram Through a Perfect Convex (Converging) Lens

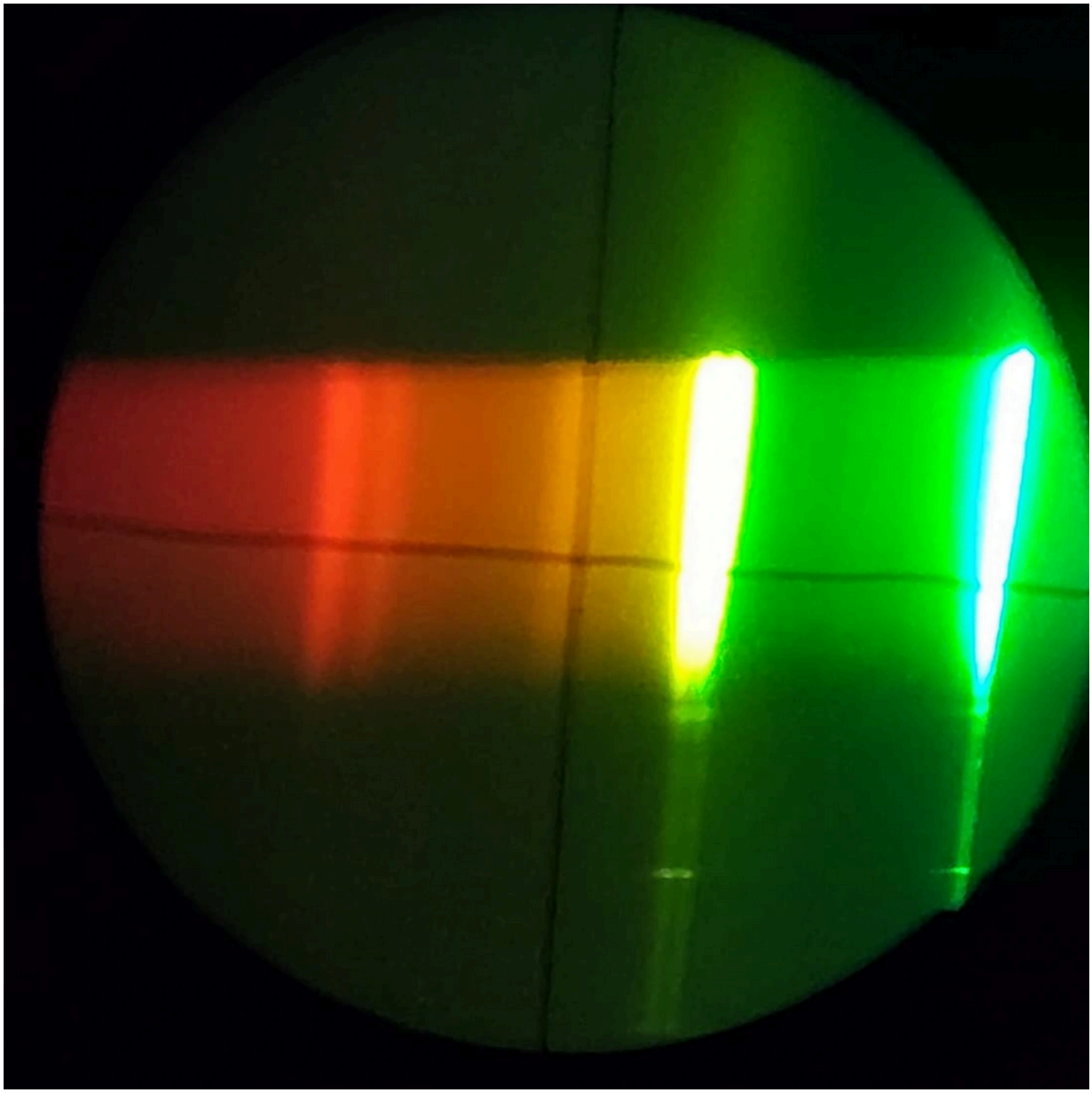

Figure C18.2: Emission Spectrum Resolved by a Diffraction Grating

Figure C18.3: NASA Researcher Aligning a Laser Device for a Laser-Powered Model Aircraft Demonstration

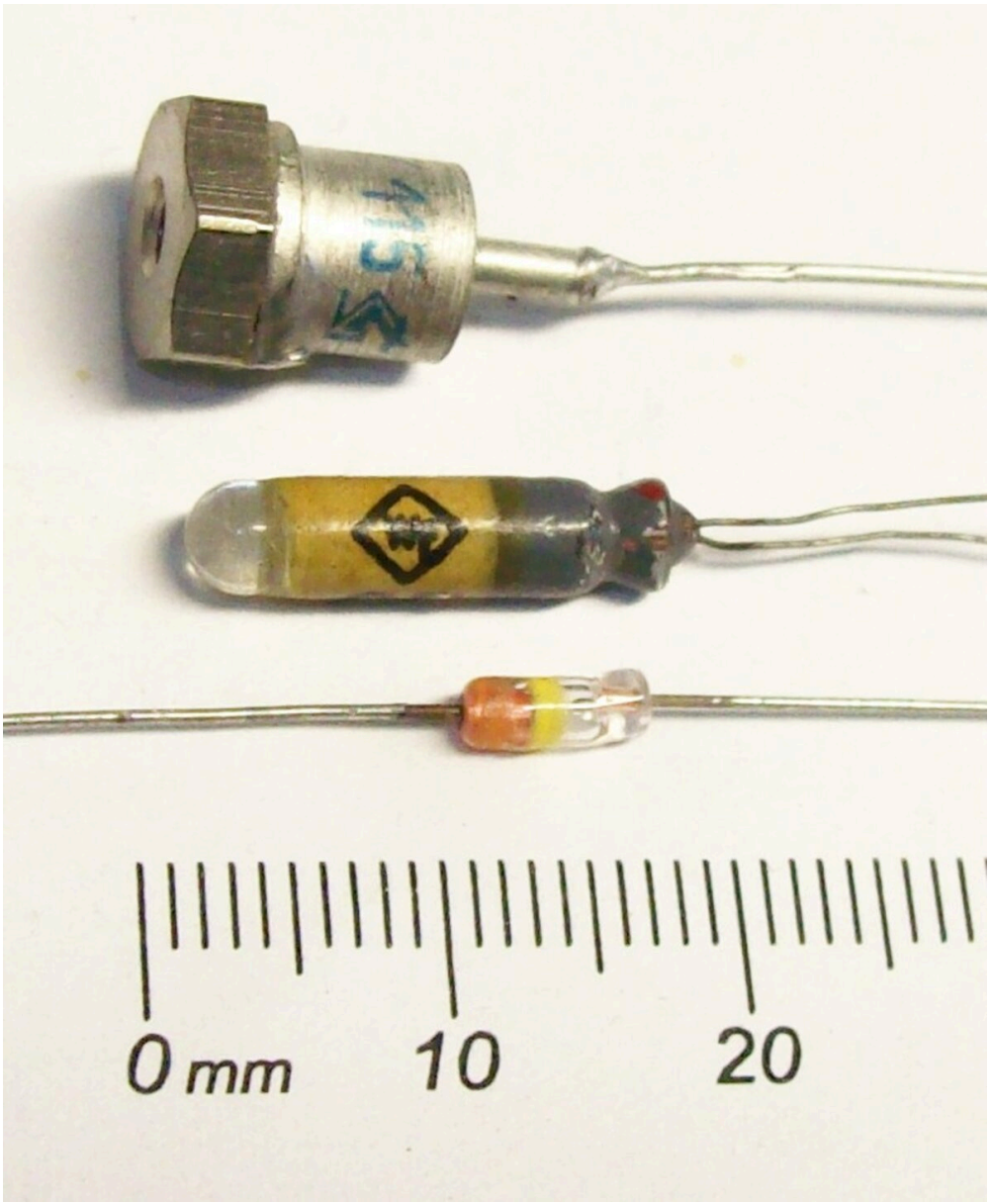

Figure C18.4: Germanium Photodiode

Figure C18.6: Hubble Space Telescope Primary Mirror — a Precision Optical System

Chapter 19: Engineering Economics

Figure C19.11: U.S. Currency — Illustrating Purchasing Power and Inflation

Appendix A: Imaginary Numbers and Phasors

Figure CA.4: Charles Proteus Steinmetz, Pioneer of the Phasor Method for AC Circuit Analysis

Appendix B: Arctangent and atan2

Appendix C: Decibels

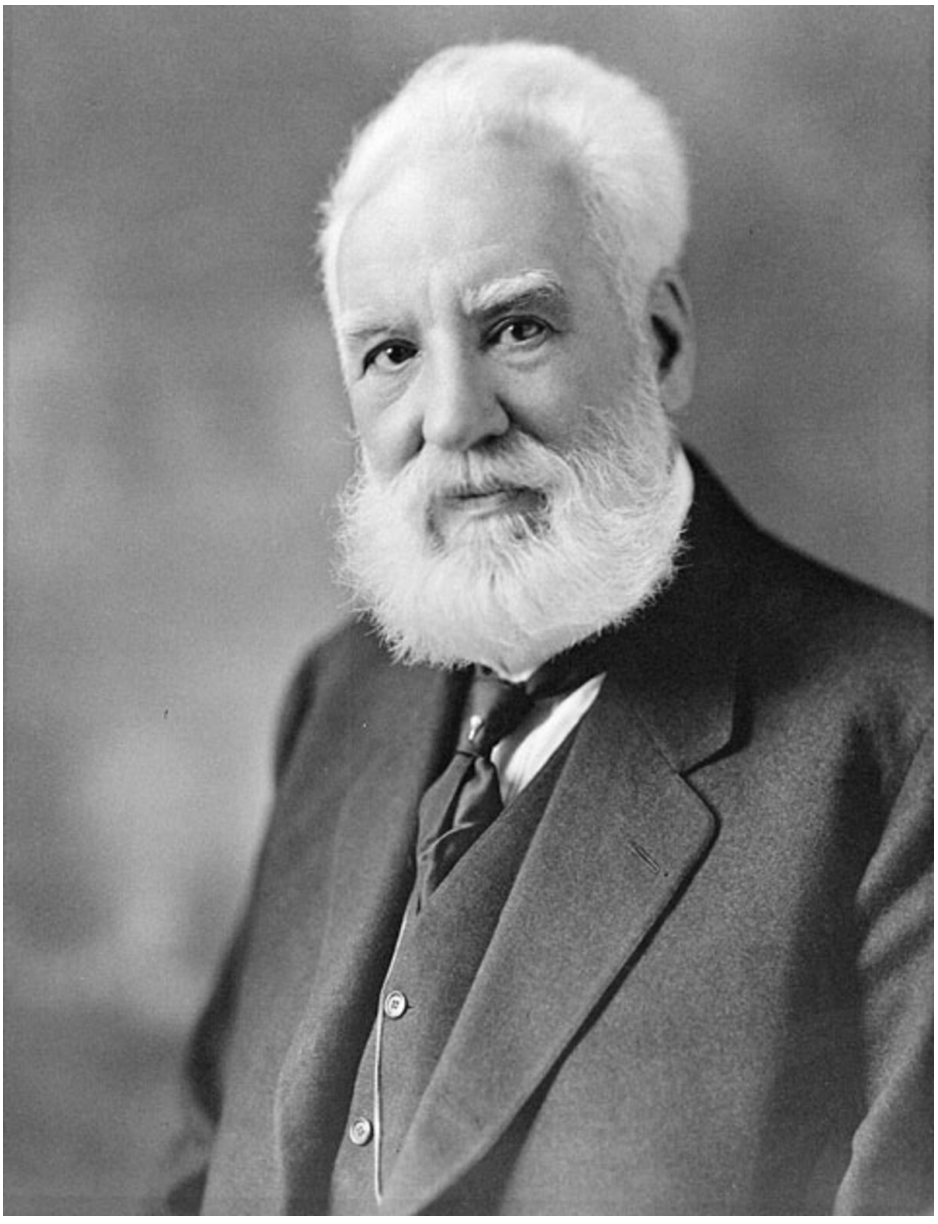

Figure CC.1: Alexander Graham Bell, Namesake of the Bel/Decibel

Appendix D: Unit Prefixes and SI Units

Figure CD.1: U.S. National Prototype Kilogram K20 — a Historical SI Base Unit Standard

Appendix E: Getting Started with Python and marimo

Appendix F: Matrix Operations

Appendix G: Op-Amp Circuit Collection